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ONLINE CERTIFICATION COURSE ON

Statistical and Machine Learning Methods: Basics and Applications

MODULE 12: Applications of Statistical and Machine Learning Methods

W12L1: K-Nearest Neighbours (KNN)

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CONCEPTS COVERED

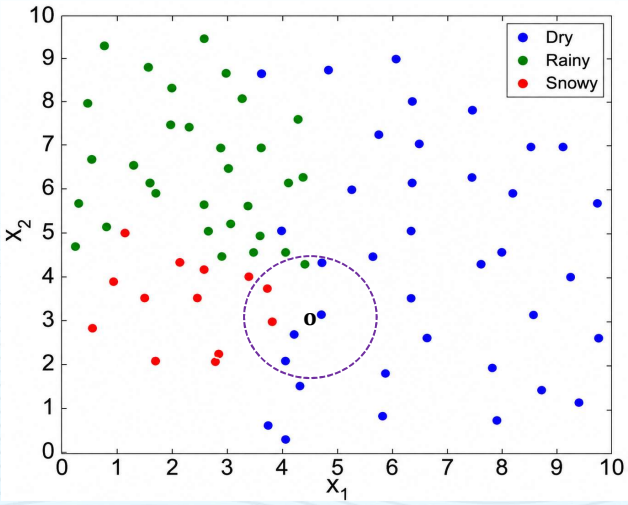
K-Nearest Neighbours (KNN) for Weather Class Forecasting

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K-Nearest Neighbours (KNN)

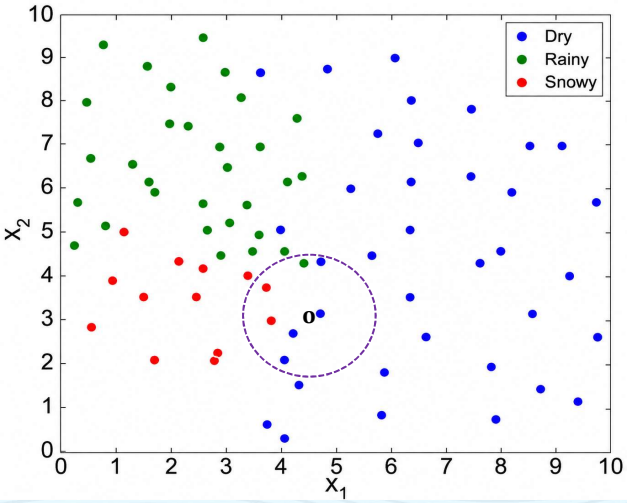
- K-Nearest Neighbours (KNN) is a relatively simpler, yet powerful supervised machine learning algorithm used for both classification and regression tasks.
- It is a non-parametric method. As a memory-based classifier, KNN stores all the training data in memory.
- The basic idea of KNN is simple: given a D-dimensional feature vector $\mathbf{x}$, one finds K feature vectors closest to $\mathbf{x}$ in the training dataset based on some distance measure (e.g. the Euclidean distance).
- For classification, the class $\hat{y}$ to assign to $\mathbf{x}$ is decided by simple voting from the classes in the K.

K-nearest Neighbours (KNN)



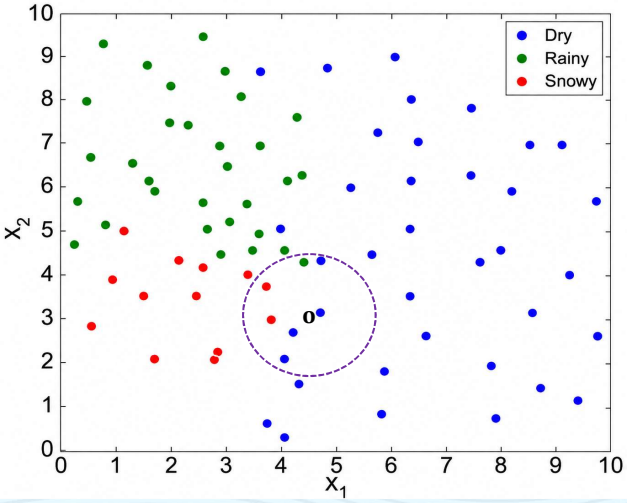
- $\mathbf{x}$ is a feature vector containing various meteorological measurements
- y is the weather class of the next day, with three classes, 'dry', '$rainy$' and '$snowy$'.

K-nearest Neighbours (KNN)



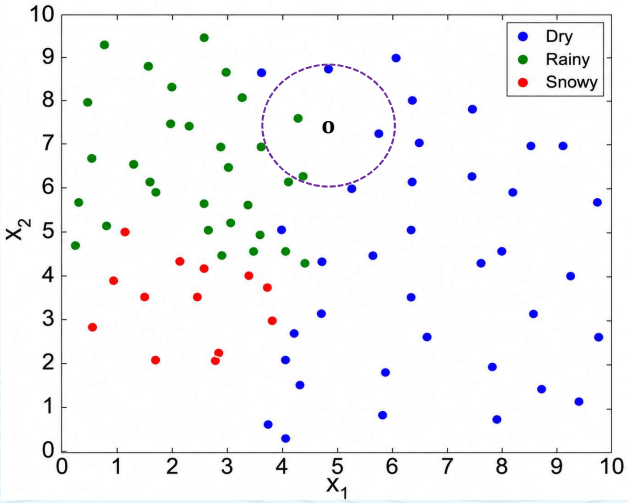
- Illustrating seven nearest neighbours (within the dashed circle) around a particular feature vector x in a 2-D feature space for the KNN classifier with $K = 7$.

K-nearest Neighbours (KNN)



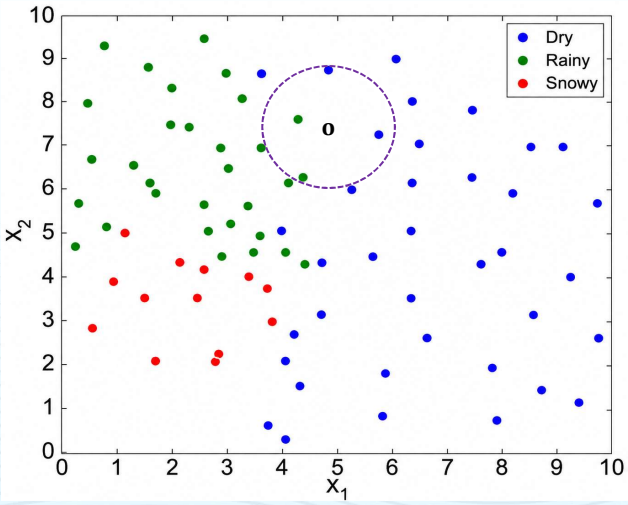
- Among the seven nearest neighbours around the point **o**, four are labelled 'dry', one 'rainy' and two 'snowy'; thus, 'dry' receives 4 votes, 'rainy' 1 and 'snowy' 2 votes, so the model output is 'dry'.

K-nearest Neighbours (KNN)



- If the nearest neighbours happen to be 2 ‘dry’, 2 ‘rainy’ and 0 ‘snowy’, we have a tie in the voting.

K-nearest Neighbours (KNN)



- To break the tie two options are there. One option is to just choose randomly among the tied top classes.
- Another option is to use the closest neighbour as the tie-breaker.

K-Nearest Neighbours (KNN) for Regression

- For regression, KNN output is the mean of the y values of the K nearest neighbours.
- More formally, for the KNN model, we can define a posterior probability

$$P(C_i|\, x) = \frac{1}{K} \sum_{j \in N_K\,(x)} \delta(y_j, C_i)$$

where the K nearest neighbours in the set $N_K\,(x)$ have class labels $\{y_j\}$ and

$$\delta(u, v) = \begin{cases} 1, & if\ u = v \\ 0, & if\ u \neq v \end{cases}$$

- The summation in the equation simply counts the number of data points among the K nearest neighbours that belong to class C_i. KNN outputs $\hat{y} = C_l$, where $P(C_l|\mathbf{x}) \geq P(C_i|\mathbf{x})$ for all $i \neq l$.

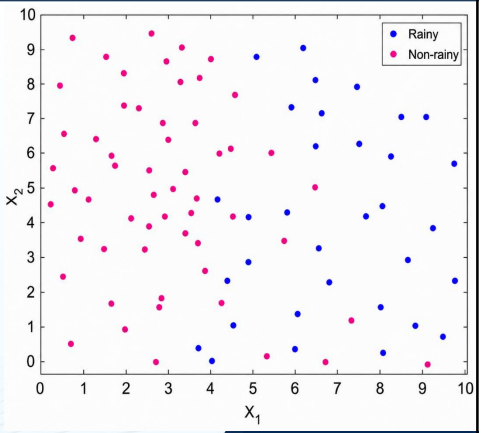
K-nearest Neighbours (KNN)

- For KNN to work well, one needs to choose an optimal value for the hyperparameter K.
- A common 'rule of thumb' is to choose $K = \sqrt{N}$, with N being the sample size of the training dataset.
- Another one is to use an ensemble of KNN models, with K = 1, 3, 5, …, K_{max}, where K_{max} is the largest odd integer $\leq \sqrt{N}$.
- Each ensemble member votes for a particular class, and the ensemble model gathers all the votes from its members and chooses the class with the most votes.
- K can also be determined through cross-validation as well.

Example-1

It is desired to forecast the weather a day whether it will be a rainy (R) or non-rainy (NR) using the rainfall magnitude on previous two days: Rainfall magnitude on previous day (t-1) is designated as x_1 and previous to previous day $(t-2)$ rainfall (x_2)

Using the data from several days from the record, the rainfall magnitudes on the previous two days and the corresponding weather condition of the current day are available.



Example-1

For demonstration, we consider following five cased from the historical records:

Neighbours	x_1	x_2	Class
1	8	2	NR
2	1	3	NR
3	4	7	R
4	6	3	R
5	7	6	R

For a new case, previous two days rainfall magnitudes are 4 mm and 1 mm, i.e., $X = (X_1, X_2) = (4,1)$.

Using the K-Nearest Neighbours (KNN) algorithm with $K = 3$, determine whether next days will be a rainy (R) or non-rainy (NR) day.

Solution

Euclidean distance is $d = \sqrt{(X_1 - x_1)^2 + (X_2 - x_2)^2}$

Calculate the distance from $X = (X_1, X_2) = (4,1)$ to neighbour 1 point event

$$d_1 = \sqrt{(4-10)^2 + (1-2)^2} = \sqrt{37} = 6.324$$

Similarly other neighbours points details are in following table

Neighbours	x_1	x_2	Class	Euclidean distance	Rank
1	8	2	NR	4.12	3
2	1	3	NR	3.61	2
3	4	7	R	6.00	5
4	6	3	R	2.83	1
5	7	6	R	5.83	4

Solution

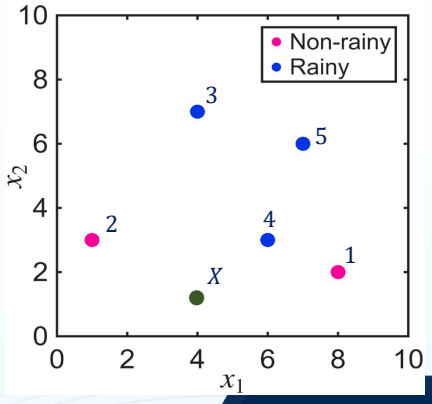
From the figure, for $K = 3$ three nearest neighbours : 4,2,1 and their classes are: R, NR and NR.

Majority voting:

a) NR: 2 (nearest neighbours 2 and 1)

b) R : 1 (nearest neighbours 4)

Hence, Class NR receives the majority vote, the predicted class of new rainfall event $X = (4,1)$ is classified as Class NR (non-rainy day).



Example-2

Neighbours	x_1	x_2	Class
1	8	2	NR
2	1	9	NR
3	4	7	R
4	6	3	R
5	7	6	R
6	3	5	R
7	5	1	NR
8	2	4	NR
9	1	10	NR
10	4	3	R

Extending Example 1, consider 10 neighbouring rainfall events as given in the following table. For a new rainfall event, $X = (X_1, X_2) = (2,9)$. Using the K-Nearest Neighbours (KNN) algorithm with $K = 4$ and Euclidean distance, determine whether the new rainfall event belongs to the rainy (R) or non-rainy (NR) class.

Solution

Euclidean distance is from the point (2,9) is in the following table

Neighbours	x_1	x_2	Class	Euclidean distance (d) $\sqrt{(X_1 - x_1)^2 + (X_2 - x_2)^2}$	Rank
1	8	2	NR	9.22	10
2	1	9	NR	1.00	1
3	4	7	R	2.83	3
4	6	3	R	7.21	8
5	7	6	R	5.83	6
6	3	5	R	4.12	4
7	5	1	NR	8.54	9
8	2	4	NR	5.00	5
9	1	10	NR	1.41	2
10	4	3	R	6.32	7

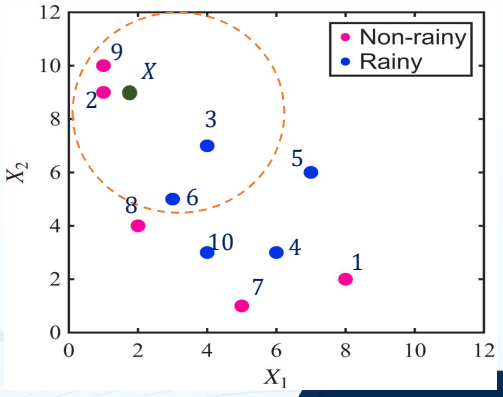
Solution

From the figure for $K = 4$, four nearest neighbours of X are: 2, 3, 6 and 9.

Majority voting:

a) NR: 2 (nearest neighbours 2 and 9)

b) R: 2 (nearest neighbours 3 and 6)



Hence, there is a tie. Both the classes receive the 2 votes. We use the nearest one as a tie break. Thus, the predicted class of next day is non-rainy.

Summary

- KNN is relatively easier to understand and simpler to implement.
- We do not need the model to be trained, when new data become available there is no need to retrain the model.
- Being non-parametric, KNN makes no assumption about the underlying distribution of the data.
- On the negative side, KNN needs to store all data in memory and can be expensive when making predictions. KNN suffers from the 'curse of dimensionality' when D, the dimension of the feature vector, grows.
- One may try to reduce D by some dimension reduction technique, such as principal component analysis.


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SWAYAM NPTEL
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Statistical and Machine Learning Methods: Basics and Applications
MODULE 12: Applications of Statistical and Machine Learning Methods
W12L2: Support Vector Regression for Monthly Streamflow Prediction
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CONCEPTS COVERED
Application of Support Vector Regression:
Prediction of Monthly Streamflow
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Importance of Streamflow Prediction

- Streamflow is one of the major hydrological variables governing the availability and movement of surface water within a river basin.
- Reliable prediction of streamflow is essential for the effective planning and management of available water resources.
- Accurate streamflow information supports several water-resources applications, including: reservoir operation, flood control and warning, planning of reservoir capacity, agriculture, hydropower generation, navigation, etc.
- Therefore, the ability to predict future streamflow conditions with reasonable accuracy is an important requirement in hydrological and water-resources decision making.

Motivation of the Study

- A variety of statistical and other modelling approaches have been developed to capture the properties of hydrological time series for reliable prediction.
- The inherent complexity of hydrological variables can limit the applicability of traditional modelling approaches.
- Machine Learning techniques, such as artificial neural networks, fuzzy logic and genetic programming have been used for modelling and prediction of hydrological variables.
- Support Vector Machines provide a kernel-based learning approach and have shown promising performance in hydrology and related applications.

Objective of the Study

- The study investigates the potential of Support Vector Regression (SVR) for prediction of monthly streamflow.
- Mahanadi River basin is considered as the study area.
- The prediction is based on the endogenous property of the streamflow time series, i.e., information contained in previous streamflow values is used to estimate the subsequent value.
- For a monthly streamflow series Q_t, the prediction framework may be expressed as:

$$Q_t = f(Q_{t-1}, Q_{t-2}, \cdots, Q_{t-p})$$

where p represents the number of previous monthly streamflow values used as predictors.

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Objective of the Study

- Four different input configurations were subsequently examined:
  - **1 input:** Q_{t-1}
  - **2 inputs:** Q_{t-1}, Q_{t-2}
  - **3 inputs:** Q_{t-1}, Q_{t-2}, Q_{t-3}
  - **4 inputs:** Q_{t-1}, Q_{t-2}, Q_{t-3}, Q_{t-4}
- The purpose was to identify the number of previous time steps that produced the best prediction performance while maintaining model parsimony.
- It is important to note that exogenous variables, such as rainfall, are not used.

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SVR and LS-SVM Framework

- SVM is a supervised Machine Learning approach that can be used for both classification and regression.
- For regression, the relationship between an input vector x and the output y is represented as:

$$y = w^T \phi(x) + b$$

where w is known as the weight vector and b the bias.

- The function $\phi(x)$ is a mostly non-linear function which maps the data into a higher, possibly infinite, dimensional feature space.

SVR and LS-SVM Framework

- The method uses equality constraints and a least-squares cost function are employed.
- The corresponding regression model is expressed as:

$$\hat{y}(x) = \sum_{i=1}^{N} \hat{\alpha}_i K(x, x_i) + \hat{b}$$

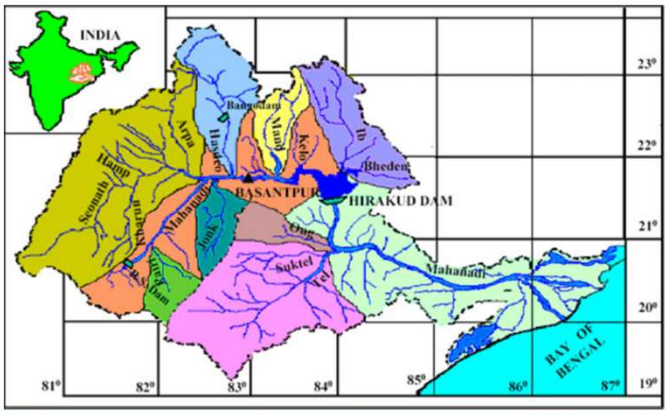
where $K(x, x_i)$ is the Kernel function.

- Here, the non-linear Radial Basis Function (RBF) kernel is adopted:

$$K(x, x_i) = exp\left[\frac{-1}{\sigma^2}\|x - x_i\|^2\right]$$

- Here, σ is the kernel function parameter of the RBF kernel.

Study Area and Streamflow Dataset



- The Mahanadi River, which encloses a drainage area of 1,32,100 sq. km, is located in the eastern part of India.
- It is one of the longest rivers in the country and drains a substantial part of peninsular India.
- The average annual rainfall in the basin is1463 mm.

Study Area and Streamflow Dataset

- For this river basin, rainfall comes predominantly from the summer monsoon (June through September).
- The streamflow data was collected from the Basantpur station for this study.
- Among these records, data for 23 years (June 1972 to May 1995) were used for calibration, and data for 9 years (June 1995 to May 2004) were used to test the model performance.

Data Normalization and Model Calibration

- The observed monthly streamflow data were normalized.
- For the ith year and j^{th} month, normalization was carried out as:

$$y_{i,j} = \frac{x_{i,j} - \mu_j}{\sigma_j}$$

where $y_{i,j}$ is the normalized value for i-th year and j-th month; $x_{i,j}$ is the observed value for i-th year and j-th month; μ_j and σ_j are the mean and standard deviation, respectively, for j-th month.

Data Normalization and Model Calibration

- The model outputs were subsequently back-transformed to the original scale.
- The model requires calibration of two parameters:
  - γ – Regularization parameter
  - σ – RBF kernel parameter
- Since their optimum values are not known beforehand, the study used a grid-search technique together with cross-validation on the training dataset.

Selection of Model Parameters and Hyperparameters

- To determine the suitable combination of γ and σ, one of the four datasets containing one lagged input vector was used as a sample calibration dataset.
- Model performance for different combinations of γ and σ was evaluated using:
  - Correlation Coefficient (CC)
  - Root Mean Square Error (RMSE)
  - Nash–Sutcliffe Efficiency (NSE)
- A higher CC and NSE and a lower RMSE indicate improved model performance.

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Selection of LS-SVM Parameters

σ	γ					
	0·1	2	4	6	8	10
0.5	0·7264	0·7091	0·7022	0·6993	0·6979	0·6971
	0·9928	0·8847	0·884	0·8841	0·8841	0·884
	0·301	0·4449	0·4458	0·4456	0·4456	0·4458
1	0·7361	0·7207	0·7141	0·7101	0·7074	0·7054
	0·9776	0·8835	0·8807	0·8804	0·8806	0·881
	0·3222	0·4464	0·4499	0·4503	0·45	0·4495
1.5	0·7391	**0·7231**	0·7206	0·7191	0·7178	0·7166
	0·9704	**0·889**	0·8847	0·8825	0·8812	0·8804
	0·3322	**0·4395**	0·4449	0·4477	0·4493	0·4503
2	0·741	0·7217	0·72	0·7194	0·7191	0·7189
	0·966	0·8924	0·8895	0·8878	0·8866	0·8855
	0·3382	0·4352	0·4389	0·441	0·4426	0·4439
2.5	0·7426	0·7204	0·7188	0·7183	0·7182	0·7181
	0·963	0·894	0·8919	0·8908	0·8899	0·8892
	0·3424	0·4331	0·4358	0·4372	0·4383	0·4392
3	0·7438	0·7193	0·7175	0·7172	0·7171	0·7172
	0·9608	0·895	0·8934	0·8925	0·8918	0·8913
	0·3453	0·4319	0·434	0·4351	0·4359	0·4366

- The paper reports the best parameter combination as: $\gamma = 2$ and $\sigma = 1.5$.
- These values were subsequently used for further model testing and validation.
- values from top to bottom in each cell are: CC, RMSE and NSE.

Selection of Endogenous Inputs

- After fixing the model parameters, the study evaluated four different cases based on the number of previous monthly streamflow values used as endogenous inputs.
- A total of 276 training datasets containing one, two, three and four previous time-step inputs were used for model development, while the remaining 102 datasets were used for validation.

No. of inputs	Performance statistics		
	RMSE	Correlation coefficient	NSE coefficient
1	0·8890	0·7230	0·4395
2	0·8716	0·7475	0·4612
3	1·0193	0·5793	0·2632
4	1·0745	0·5232	0·1812

- The two-input case produced the best performance, with the lowest RMSE and highest CC and NSE.
- Thus, for this streamflow series, the optimum endogenous input structure was:

$$Q_t = f(Q_{t-1}, Q_{t-2})$$

Comparison with other Models

- To assess the relative prediction skill of SVR, its performance was compared with the traditional Box-Jenkins models, which also uses the endogenous characteristics of the streamflow time series.
- Different types of Box–Jenkins models are known, such as auto-regressive (AR) and auto-regressive moving average (ARMA) or ARIMA.
- The model order and parameter values depend on the statistical properties of time series.
- For example, an AR model is a linear regression of the current value of the series against one or more prior values of the series.
- Thus, the AR model for univariate time series can be expressed as: $x_t = \delta + \phi_1 x_{t-1} + \phi_2 x_{t-2} + \cdots + \phi_p x_{t-p} + e_t$

Comparison with other Models

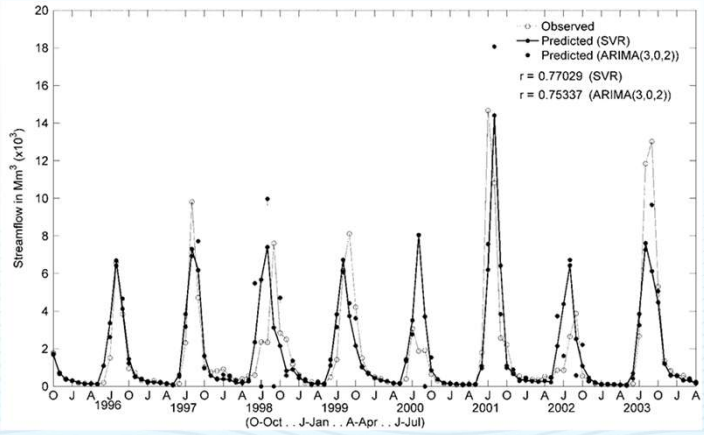
where $\delta = \left(1 - \sum_{i=1}^{0} \phi_i\right)$, with μ denoting the process mean, x_t is the observed value and e_t is white noise at time t. The value of p is called the order of the AR model.

- The autocorrelogram and partial autocorrelogram of the streamflow series were examined, based on which two models were tentatively selected: ARIMA (5,0,1) and ARIMA (3,0,2).
- ARIMA(3,0,2) showed better performance than ARIMA(5,0,1).
- However, the performance of SVR is even better than both the ARIMA models.

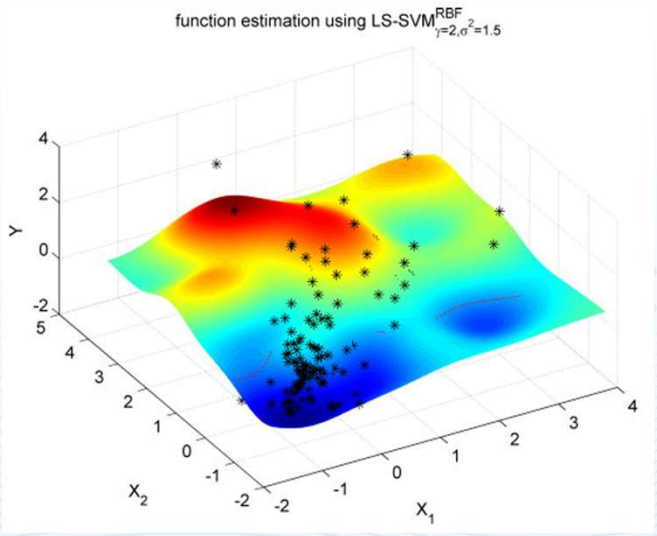
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Comparison of SVR and ARIMA Predictions

- SVR therefore provided better correspondence between observed and predicted monthly streamflow.



Comparison of SVR and ARIMA Predictions



- The superior performance of SVR can be explained through its ability to map the input data into a higher-dimensional feature space.
- For the optimum two-input case, the relationship among the two input streamflow values and the predicted output can be represented as a three-dimensional input–output surface.

Comparison of SVR and ARIMA Predictions

- In the SVR model, the fitted surface exhibits a complex nonlinear form containing different peaks and troughs.
- The degree of smoothness and the degree of fitting through observations are competing properties controlled by the LS-SVM parameters γ and σ.
- This nonlinear feature-space representation allows SVR to capture complex characteristics of the streamflow time series.
- In contrast, the corresponding fitted surface in the conventional ARIMA approach is linear for the two-input case.
- Thus, for a linear time series, the prediction performances of SVR and ARIMA may be comparable.
- When nonlinearity exists, SVR can provide improved prediction because that nonlinearity can be represented through the feature space.

Interpretation of Prediction Errors

- Although the SVR model showed better overall performance than the Box–Jenkins models, some limitations were observed during the testing period.
- The correlation coefficient between observed and predicted streamflow using SVR was approximately 0.77.
- However, some monsoon peak streamflow values were not adequately predicted, particularly in years such as 2000 and 2003.
- It is also observed that there is a lag between some observed and predicted peak streamflow peaks.
- The authors attribute this partly to the fact that the model uses only previous streamflow values as inputs.
- It indicates that information contained in the streamflow time series alone may not be sufficient to represent all variations in monthly streamflow.

CONCLUSIONS

- The SVR model shows marginally better performance for monthly streamflow prediction at Basantpur station on the Mahanadi River.
- Considering model parsimony, the two-input case provided the best performance. SVR showed better prediction performance than the Box–Jenkins models considered in the study.
- Some peak streamflow values were not adequately captured using only endogenous streamflow information.
- Reference:

Maity R., P. P. Bhagwat and A. Bhatnagar, (2010), Potential of Support Vector Regression for Prediction of Monthly Streamflow using Endogenous Property, Hydrological Processes, Wiley, 24(7), 917-923, https://onlinelibrary.wiley.com/doi/abs/10.1002/hyp.7535.


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Statistical and Machine Learning Methods: Basics and Applications

MODULE 12: Applications of Statistical and Machine Learning Methods

W12L3: Global Climate Fields to Regional Precipitation Anomalies

Prof. Rajib Maity
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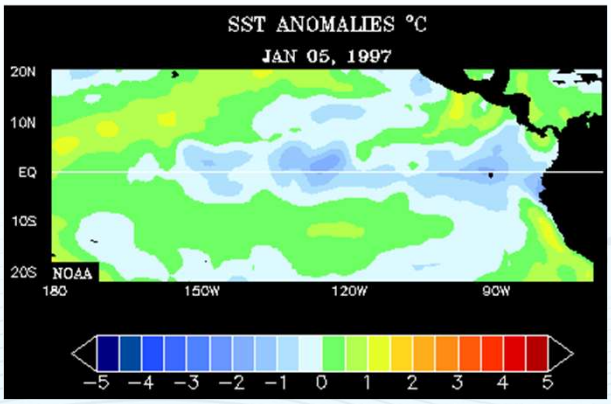
CONCEPTS COVERED

Application of PCA and Support Vector Machines:

Global Climate Fields to Regional Precipitation Anomalies

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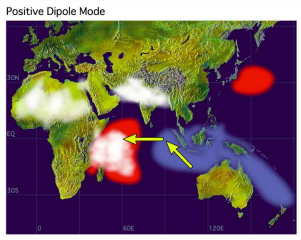
El Niño-Southern Oscillation (ENSO)



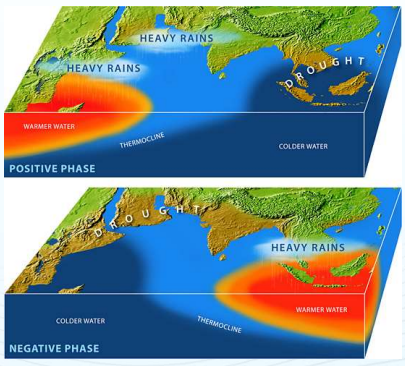
- *El Niño-Southern Oscillation (ENSO)* is a large-scale coupled oceanic-atmospheric circulation over tropical Pacific Ocean.
- Attempts were made to forecast hydrologic variables, like rainfall, streamflow, etc. using ENSO information all over the world.

Source: NOAA

Indian Ocean Dipole (IOD) mode

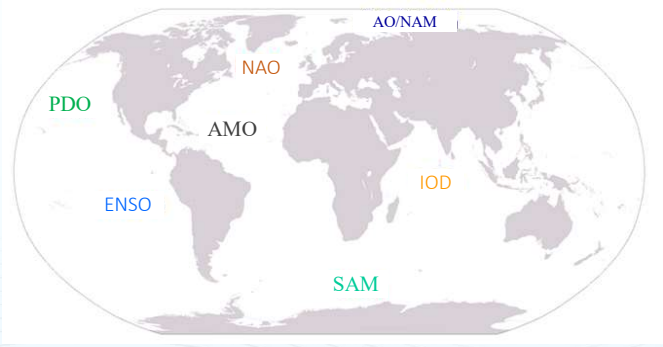






Sources: UCAR & JAMSTEC

Most heavily researched atmospheric patterns



- AMO: Atlantic Multidecadal Oscillation
- PDO: Pacific Decadal Oscillation
- SAM: Southern Annular Mode
- ENSO: El Niño-Southern Oscillation
- IOD: Indian Ocean Dipole
- NAO: North Atlantic Oscillation
- AO/NAM: Arctic Oscillation/ Northern Annular Mode

Uncovering Global Climate Fields causing Local Extremes

- The concurrent influence of large-scale, coupled oceanic-atmospheric circulation patterns is established to have an effect on hydrologic variability across the world.
- El Niño Southern Oscillation (ENSO) and Indian Ocean Dipole (IOD) are in particular, important for Indian hydroclimatology.
- It may be possible that local (continental-scale) hydrologic above/below normal events are influenced by a concurrent effect of different climate variables from across the globe rather than just a few established teleconnection patterns.
- Is there any global pattern for hydrological extremes and its potential to predict the above/below normal events? We hypothesize that a specific regional hydrological phenomenon is, in general, influenced by the concurrent effect of a specific global climate pattern (GCP).
- We explored it for the entire Indian mainland using Machine Learning.

Study Area and Data

- All India rainfall from India Meteorological Department (IMD) for the period 1959-2010.
- Monthly global fields of different climate variables from National Oceanic an Atmospheric Administration (NOAA) for the period 1958-2010 (Reanalysis I).

Name of the variable	Abbreviation	Spatial Resolution	Units
Sea Surface Temperature	SST	2.0º x 2.0º	ºC
Surface Pressure	SP	2.5º x 2.5º	millibars
Air Temperature	AT	2.5º x 2.5º	ºC
Wind Speed	WS	2.5º x 2.5º	m/s
Total Precipitable Water	TPW	2.5º x 2.5º	kg/m^2
Relative Humidity	RH	2.5º x 2.5º	%

Quantification of local extremes through SPAI

- Standardized precipitation anomaly index (SPAI).
- This index retains the benefit of the widely used standardized precipitation index (SPI). However, it has two methodological differences.
- In the computation of SPAI, firstly, the anomaly series of precipitation is used instead of the absolute precipitation series and, secondly, a single probability distribution is fitted instead of 12 different distributions used in SPI.
- The anomalies of precipitation are given by:

$$y_{ij} = \frac{x_{i,j} - \bar{x}_j}{s_j}$$

Standardized Precipitation Anomaly Index (SPAI)

- After obtaining the anomalies, a *t-location-scale probability distribution* is fitted to the entire anomaly series. The pdf of *t-location-scale probability distribution* is given by:

$$f(y|\mu,\sigma,\nu) = \frac{\Gamma\left(\frac{\nu+1}{2}\right)}{\sigma\sqrt{\nu\pi}\,\Gamma\left(\frac{\nu}{2}\right)}\left[\nu + \left(\frac{y-\mu}{\sigma}\right)^2\right]^{-\left(\frac{\nu+1}{2}\right)} \quad ; -\infty < x < \infty$$

  where μ, σ and ν are the location, scale ($\sigma > 0$) and shape ($\nu > 0$) parameters respectively.

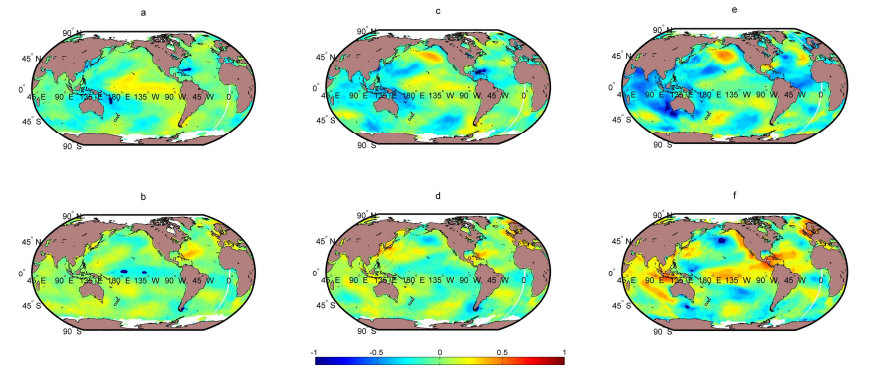
- Range of SPAI values is $-\infty$ to $+\infty$, with negative values indicating dryness and positive values indicating wetness.
- SPAI $\leq 0.8 \longrightarrow$ Dry Event
- SPAI $> 0.8 \longrightarrow$ Wet Event

Chanda K., and **R. Maity**, (2015), Meteorological Drought Quantification with Standardized Precipitation Anomaly Index (SPAI) for the Regions with Strongly Seasonal and Periodic Precipitation, *Journal of Hydrologic Engineering*, ASCE, **20**(12), 06015007-1 – 06015007-8, https://doi.org/10.1061/(ASCE)HE.1943-5584.0001236.

Identification and Characterization of Global Climate Pattern

- The extreme events (i.e., dry and wet) in the target region (India) are picked out.
- For each dry or wet event of temporal scale 3 months, 3-month standardized (with long-term standard deviation) anomalies of the climate variables are obtained at 3-months lag at all grids across the globe.
- The mean climate anomalies of all *dry* events are determined by event-wise averaging of the concerned climate anomaly over each grid.
- Similarly, gridwise mean anomalies of all wet events are also computed for each climate variable.
- Maps showing the global field of anomaly differences during dry and wet events are plotted for inspection.

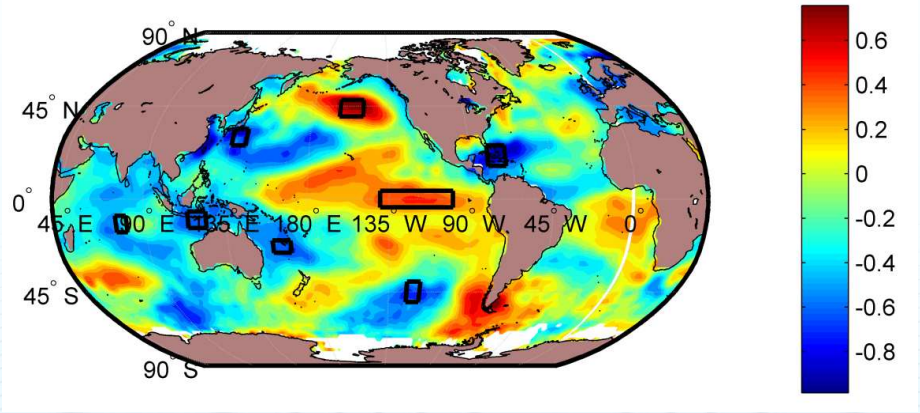
SST Zones with Opposite Anomalies during Dry and Wet Events



Mean SST fields during dry (SPAI ≤ 0.8) and wet (SPAI > 0.8) events, respectively, at (a) and (b) 1-month, (c) and (d) 3-month, (e) and (f) 6-month temporal scales.

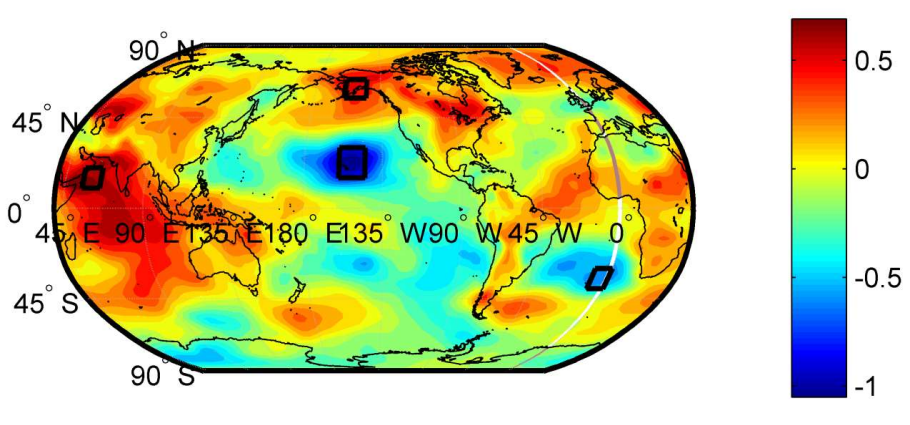
The range of the color bar is fixed between 1 and −1 for ease of comparison. The zones with opposite features gradually improve with respect to their extent and crispness with the increase in the temporal scale of SPAI.

SST Zones with Opposite Anomalies during Dry and Wet Events



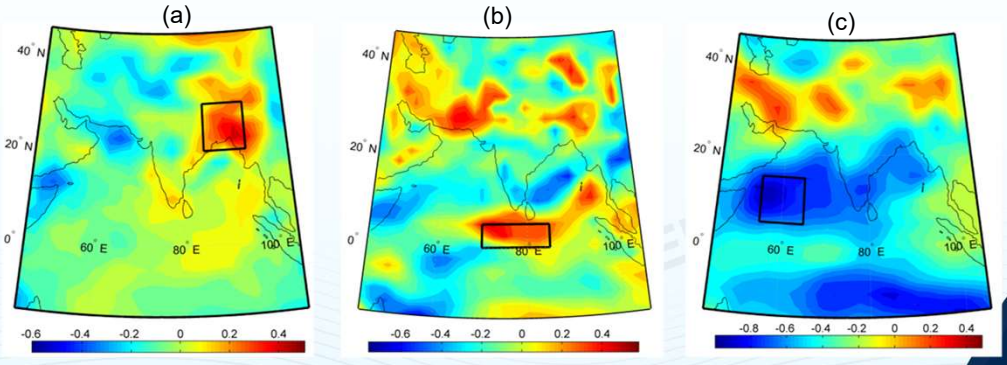
Differences in mean SST anomalies during dry events and wet events at 3-month temporal scale. Possible representative zones are shown in rectangles.

SP Zones with Opposite Anomalies during Dry and Wet Events



Differences in mean SP anomalies during dry events and wet events at 3-month temporal scale. Possible representative zones are shown in rectangles.

AT, WS and TPW Zones with Opposite Anomalies during Dry and Wet Events



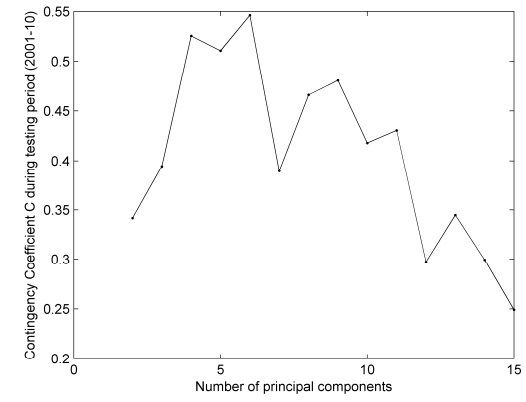
Differences in mean anomalies during dry and wet events at 3-month temporal scale for (a) AT (b) WS and (c) TPW. Possible representative zones are shown in rectangles

Potential of the GCP

- In all, 15 different variables, each designated by a climate anomaly from a distinct zone on the globe, are found to form a potential global climate signal pool, which can be designated as the GCP for Indian rainfall.

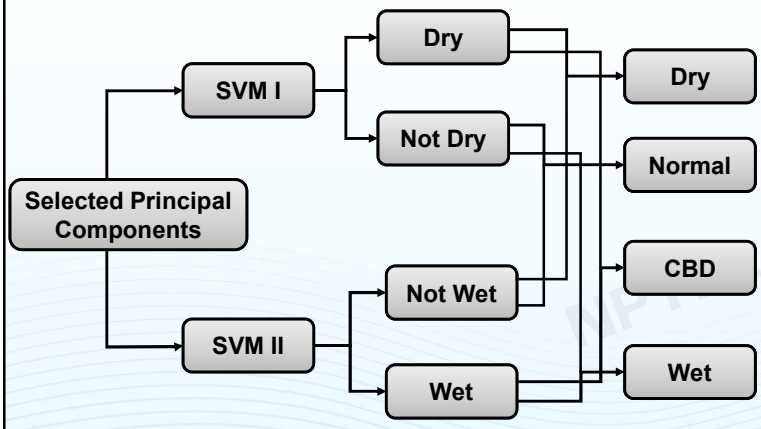
- Dimensionality reduction of extensive GCP
- Prediction of extreme precipitation events
- Evaluation of the potential of GCP in comparison to previously known hydroclimatic indicators

Dimensionality reduction of extensive GCP



- High dimensionality could hinder the effective extraction of useful information in the prediction process.
- By applying PCA on the n-dimensional input set, a set of principal components is obtained in the order of gradually reducing explained variability.
- It is found that the optimum number of principal components is six which gives the best performance (in terms of contingency coefficient C which is explained later) during testing period.

Prediction Model of Extreme Precipitation Events



- Two Support Vector Machine (SVM) models are applied.
- SVM-I model is trained for classification into 2 categories – *dry* and *not dry*, SVM-II model is trained for classification into – *not wet* and *wet.*

- After training the two SVMs, they are used for the prediction of precipitation extremes for both model development and testing period (CBD→ Cannot Be Determined)

Evaluation of the potential of GCP

- The potential of the GCP may be assessed by examining the ability of the model to correctly predict the category of the precipitation events using the optimum number of principal components of GCP as input.
- A contingency table is prepared and the performance is evaluated in terms of Contingency Coefficient C.

$$C = \sqrt{\frac{Q}{Q+N}}$$

where Q is a statistic that tests the null hypothesis that there is no association between observed and predicted categories and N is the number of samples.

- Higher values of C indicate better association between observed and predicted categories.

Contingency Table for ENSO and IOD only

	Development period (1959–2000)			Testing period (2001–2010)		
Observed category	Predicted category					
	Dry	Normal	Wet	Dry	Normal	Wet
Dry	47	6	26	1	0	18
Normal	137	29	148	4	2	71
Wet	31	7	73	0	2	22
Q	22.632			3.821		
v	4			4		
p value	1.5×10^{-4}			0.43		
C	0.207			0.176		
C_{max}	0.817			0.817		
C/C_{max}	0.253			0.215		

Contingency Table for GCP

	Development period (1959–2000)			Testing period (2001–2010)		
Observed category	Predicted category					
	Dry	Normal	Wet	Dry	Normal	Wet
Dry	53	5	17	12	1	5
Normal	122	39	140	5	3	66
Wet	22	11	68	0	3	20
Q	45.400			48.865		
v	4			4		
p value	3.2×10^{-9}			6.2×10^{-10}		
C	0.295			0.546		
C_{max}	0.817			0.817		
C/C_{max}	0.361			0.668		

Chanda K., and **R. Maity**, (2015), Uncovering Global Climate Fields Causing Local Precipitation Extremes, *Hydrological Sciences Journal*, **61**(10), 1227-1237, DOI: 10.1080/02626667.2015.1006232


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Statistical and Machine Learning Methods: Basics and Applications

MODULE 12: Applications of Statistical and Machine Learning Methods

W12L4: Potential of Deep Learning in Drought Assessment

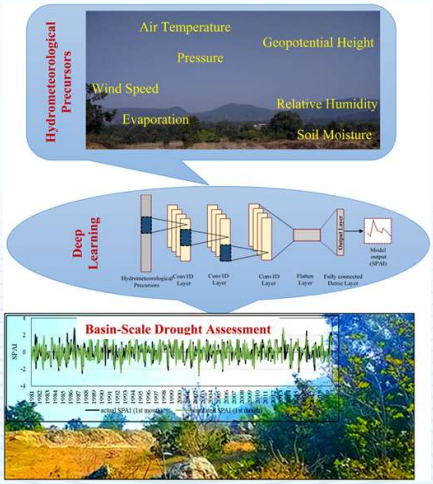
Prof. Rajib Maity
Professor, Department of Civil Engineering
Chairperson, School of Water Resources
Indian Institute of Technology Kharagpur
Kharagpur – 721302, West Bengal

CONCEPTS COVERED

Potential of Deep Learning in Drought Assessment by Extracting Information from Hydrometeorological Precursors

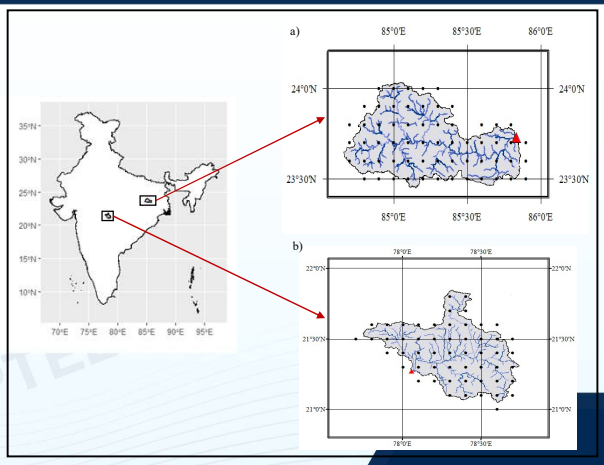
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Potential of Deep Learning in Drought Assessment by Extracting Information from Hydrometeorological Precursors



Study Area

- Location of study basins with approximate drainage network:
  - a) Damodar River Basin upto Tenughat Dam (TRB).
  - b) Wardha River Basin upto Upper Wardha Dam (WRB).
- The basin outlets are shown as red triangles and the black dots designate the grid points (0.1°x0.1°) utilized for analysis.

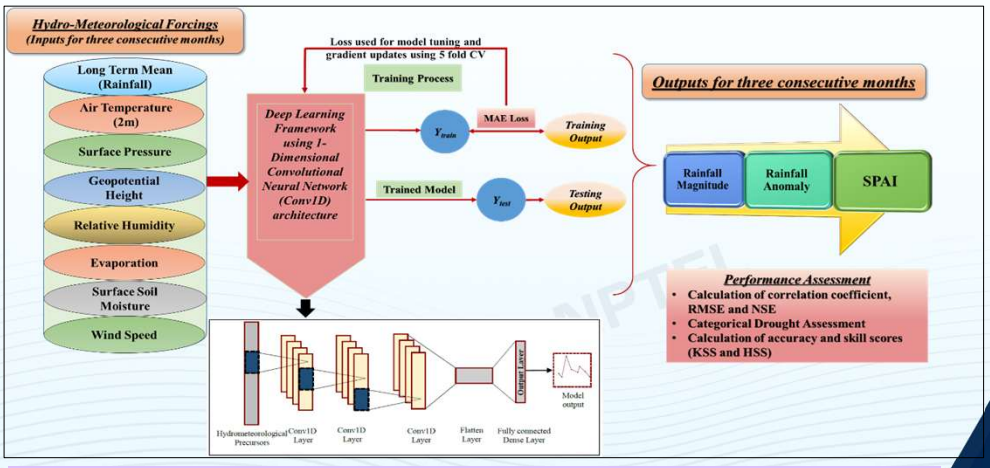


Data Used

Details of the meteorological dataset used in the study for a period of 1981-2019.

Dataset	Variables	Spatial Resolution	Vertical/Pressure Level	Units (ERA5)	Units (Converted)
ERA5-Land monthly averaged data	Total precipitation	0.1°×0.1°	Surface	m	mm
	2-m temperature	0.1°×0.1°	2-m above surface	K	°C
	Pressure	0.1°×0.1°	Surface	Pa	kPa
	Evaporation	0.1°×0.1°	Surface	m (of water equivalent)	mm
	Volumetric soil water	0.1°×0.1°	Sub-Surface (0-7 cm)	m^3/m^3	--
	10-m u-wind	0.1°×0.1°	10-m above surface	m/s	--
	10-m v-wind	0.1°×0.1°	10-m above surface	m/s	--
ERA5 monthly averaged data on pressure levels	Geopotential height	0.25°×0.25°	1000 hPa	m^2/s^2 (Geopotential)	m (by dividing geopotential by gravitational constant)
	Relative humidity	0.25°×0.25°	1000 hPa	Percent (%)	--

Methodology Framework



Methodological flowchart along with a schematic diagram showing different layers of Conv1D model architecture

Reference: Rajib Maity, Mohd Imran Khan, Subharthi Sarkar, Riya Dutta, Subhra Sekhar Maity, Manali Pal, Kironmala Chanda; Potential of Deep Learning in drought assessment by extracting information from hydrometeorological precursors. Journal of Water and Climate Change 1 September 2021; 12 (6): 2774–2796. doi: https://doi.org/10.2166/wcc.2021.062.

Drought Characterization for Indian Hydroclimatology

- Standardized Precipitation Anomaly Index (SPAI) is a generalized anomaly-based index for characterizing meteorological drought in monsoon dominated climatology (Chanda and Maity, 2015; Das and Chanda, 2020; Maity et al., 2020; Makokha et al., 2016; Monish and Rehana, 2020).
- As the rainfall in India is strongly seasonal due to its monsoon dominated climatology, the SPAI is used as the drought characterization index.

Categorical Classification of Drought

State	Description	Drought Index Range
D2	Extreme drought	SPAI < -2
D1	Severe drought	$-2 \leq$SPAI < -1.5
D0	Moderate drought	$-1.5 \leq$ SPAI < -1
N	Near normal	$-1 \leq$ SPAI ≤ 1
W0	Moderately wet	$1 <$ SPAI ≤ 1.5
W1	Severe wet	$1.5 <$ SPAI ≤ 2
W2	Extremely wet	SPAI > 2

Data Preparation

- Entire data preparation and handling are carried out in the scientific python development environment (Spyder) notebook. The steps involved in data preparation are as follows:
- Spatially averaging of the gridded data sets across the basins by using area weightage method.
- Conversion of the data to monthly scale honoring their units
- Fold division (5 fold in this study)
- Train and Test split (80% and 20% respectively)
- Arrangement of the data as per model requirement

Note: The input and target data is arranged in a manner that proposed DL model utilizes seven hydrometeorological precursors to simulate three consecutive months rainfall values. Thereafter outside the DL domain from simulated rainfall values SPAI values are computed.

Model Development

One-dimensional Convolutional Neural Network (Conv1D)

- The proposed Conv1D model is developed in the Spyder notebook using Keras library built on top of TensorFlow.
- It is of a sequential type that can be used for simulation/prediction problems using one or multiple time series.
- Several combinations of layers needs to be evaluated with different sets of model hyper-parameters (such as filter size, number of filters, dropout percentage, number of hidden layers, such as batch size, activation function, number of epochs, learning rate, loss function, momentum and decay rate) to ascertain the best possible model configuration.
- The finalized architecture of the Conv1D model used in this study comprises of five layers.

Model Development

The finalized architecture of the proposed model comprises of five layers. The configuration of these layers are:

S.No.	Layer Types	Model Parameters					
		Activation Function	No. of Filters/ Memory cells	Kernel Size	Strides	Dropout Size	No. of Nodes
1	Conv1D Layer	ReLU	124	1	1	---	---
2	Conv1D Layer	ReLU	124	1	1	---	---
3	Conv1D Layer	ReLU	124	1	1	---	---
4	Flatten Layer	---	---	---	---	---	---
5	Fully Connected Dense Layer	ReLU	---	---	---	---	3

Methodology

Comparison with other approaches

- Support Vector Regression (SVR) with RBF Kernel: $C = 100$ and $\gamma = 1 \times 10^{-5}$

Model Performance Evaluation Criteria

- Root Mean Squared Error (RMSE)
- Coefficient of correlation (r) and
- Nash–Sutcliffe Efficiency (NSE)

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Methodology

Performance of the proposed Conv1D model in simulating the drought category

Typical Contingency table

Observed	Simulated				
	S_1	S_2	…	S_c	Sum
O_1	n_{11}	n_{21}	…	n_{c1}	NO_1
O_2	n_{12}	n_{22}	…	n_{c2}	NO_2
:	:	:	…	:	:
O_c	n_{1c}	n_{2c}	…	n_{cc}	NO_c
Sum	NS_1	NS_2	…	NS_c	N

Mathematical Formulations

$$Accuracy = \frac{1}{N}\sum_{i=1}^{k} n_{ii}$$

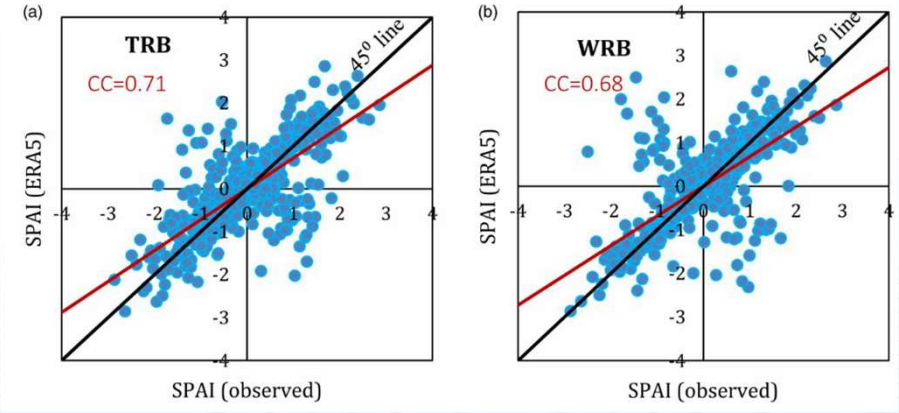
$$HSS = \frac{\frac{1}{N}\sum_{i=1}^{k} n_{ii} - \frac{1}{N^2}\sum_{i=1}^{k} NO_i NS_i}{1 - \frac{1}{N^2}\sum_{i=1}^{k} NO_i NS_i}$$

$$KSS = \frac{\frac{1}{N}\sum_{i=1}^{k} n_{ii} - \frac{1}{N^2}\sum_{i=1}^{k} NO_i NS_i}{1 - \frac{1}{N^2}\sum_{i=1}^{k} NO_i^2}$$

Heidke Skill Score (HSS) and Kuipers Skill Score(KSS)

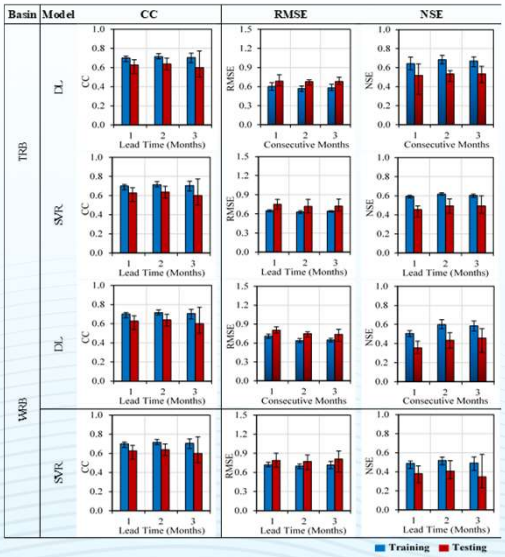
Reference: Rajib Maity, Mohd Imran Khan, Subharthi Sarkar, Riya Dutta, Subhra Sekhar Maity, Manali Pal, Kironmala Chanda; Potential of Deep Learning in drought assessment by extracting information from hydrometeorological precursors. Journal of Water and Climate Change 1 September 2021; 12 (6): 2774–2796. doi: https://doi.org/10.2166/wcc.2021.062.

Results and Discussions



Scatter plots between SPAI values calculated from observed precipitation data and that from ERA5 reanalysis precipitation data for (a) TRB and (b) WRB study basins.

Results and Discussions



Average CC, RMSE and NSE, along with range bars at the top show the maximum to minimum value, across different folds for drought category identification for both the study basins using Deep Learning (DL) and Support Vector Regression (SVR) models for 1st, 2nd and 3rd months.

Results and Discussions

Fold-wise comparison of performance metrics between observed and simulated SPAI during training (trn) and testing (tst) period obtained by Deep Learning (DL) and Support Vector Regression (SVR) models for 1st, 2nd and 3rd months for the TRB study basin

Study Basin: TRB

Fold	Model	Correlation Coefficient						RMSE						NSE					
		1st month		2nd month		3rd month		1st month		2nd month		3rd month		1st month		2nd month		3rd month	
		trn	tst	trn	tst	trn	tst	trn	tst	trn	tst	trn	tst	trn	tst	trn	tst	trn	tst
Fold 1	DL	0.82	0.80	0.83	0.77	0.83	0.79	0.58	0.61	0.57	0.66	0.58	0.63	0.67	0.64	0.68	0.57	0.67	0.61
	SVR	0.77	0.70	0.80	0.69	0.78	0.69	0.64	0.83	0.61	0.83	0.63	0.83	0.59	0.49	0.63	0.48	0.60	0.48
Fold 2	DL	0.83	0.74	0.84	0.76	0.83	0.76	0.58	0.66	0.55	0.63	0.57	0.64	0.68	0.50	0.71	0.55	0.68	0.57
	SVR	0.74	0.67	0.77	0.73	0.75	0.65	0.64	0.67	0.60	0.62	0.63	0.71	0.58	0.43	0.61	0.53	0.58	0.41
Fold 3	DL	0.81	0.70	0.82	0.74	0.81	0.75	0.66	0.78	0.61	0.69	0.64	0.70	0.58	0.32	0.64	0.46	0.61	0.44
	SVR	0.78	0.63	0.80	0.65	0.79	0.65	0.66	0.75	0.64	0.72	0.65	0.71	0.61	0.38	0.63	0.42	0.62	0.42
Fold 4	DL	0.82	0.77	0.84	0.77	0.84	0.74	0.64	0.70	0.58	0.71	0.57	0.75	0.58	0.57	0.66	0.56	0.67	0.50
	SVR	0.78	0.71	0.78	0.70	0.78	0.75	0.65	0.76	0.65	0.77	0.65	0.71	0.60	0.49	0.61	0.48	0.60	0.55
Fold 5	DL	0.85	0.77	0.86	0.74	0.85	0.75	0.54	0.67	0.52	0.70	0.54	0.69	0.71	0.57	0.73	0.53	0.71	0.55
	SVR	0.77	0.70	0.78	0.75	0.79	0.78	0.66	0.73	0.65	0.67	0.65	0.65	0.60	0.49	0.61	0.57	0.61	0.60
Average	DL	0.83	0.76	0.84	0.76	0.83	0.76	0.60	0.69	0.57	0.68	0.58	0.68	0.64	0.52	0.68	0.53	0.67	0.53
	SVR	0.77	0.68	0.79	0.71	0.78	0.70	0.65	0.75	0.63	0.72	0.64	0.72	0.59	0.45	0.62	0.49	0.60	0.49

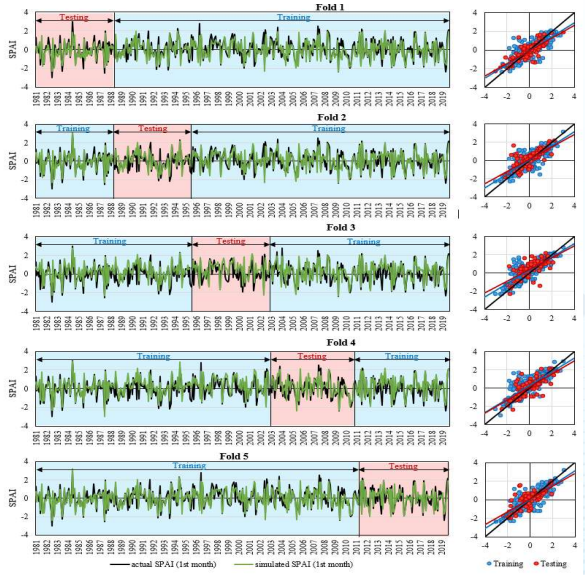
Results and Discussions

Fold-wise comparison of performance metrics between observed and simulated SPAI during training (trn) and testing (tst) period obtained by Deep Learning (DL) and Support Vector Regression (SVR) models for 1st, 2nd and 3rd months for the WRB study basin.

Study Basin: WRB

Fold	Model	Correlation Coefficient						RMSE						NSE					
		1st month		2nd month		3rd month		1st month		2nd month		3rd month		1st month		2nd month		3rd month	
		trn	tst	trn	tst	trn	tst	trn	tst	trn	tst	trn	tst	trn	tst	trn	tst	trn	tst
Fold 1	DL	0.75	0.67	0.77	0.73	0.76	0.69	0.69	0.82	0.65	0.76	0.67	0.82	0.51	0.42	0.56	0.52	0.53	0.45
	SVR	0.72	0.63	0.75	0.62	0.75	0.56	0.69	0.84	0.66	0.85	0.66	0.91	0.51	0.39	0.55	0.39	0.56	0.31
Fold 2	DL	0.72	0.68	0.79	0.72	0.78	0.77	0.74	0.75	0.64	0.69	0.65	0.63	0.48	0.34	0.61	0.44	0.59	0.56
	SVR	0.72	0.65	0.69	0.68	0.73	0.77	0.72	0.72	0.70	0.65	0.70	0.61	0.51	0.41	0.53	0.52	0.53	0.58
Fold 3	DL	0.73	0.68	0.77	0.71	0.79	0.75	0.73	0.81	0.67	0.77	0.67	0.73	0.47	0.35	0.56	0.40	0.56	0.46
	SVR	0.66	0.63	0.69	0.62	0.65	0.55	0.76	0.80	0.73	0.81	0.77	0.85	0.43	0.36	0.47	0.34	0.41	0.26
Fold 4	DL	0.75	0.66	0.81	0.68	0.81	0.66	0.70	0.78	0.61	0.75	0.62	0.78	0.53	0.30	0.65	0.35	0.64	0.31
	SVR	0.70	0.68	0.72	0.70	0.68	0.62	0.73	0.68	0.72	0.68	0.76	0.75	0.49	0.46	0.51	0.47	0.46	0.36
Fold 5	DL	0.76	0.65	0.80	0.70	0.79	0.73	0.67	0.86	0.62	0.78	0.62	0.74	0.54	0.36	0.61	0.47	0.61	0.52
	SVR	0.70	0.54	0.73	0.58	0.71	0.50	0.70	0.90	0.68	0.87	0.70	0.93	0.49	0.28	0.53	0.33	0.50	0.23
Average	DL	0.74	0.67	0.79	0.71	0.79	0.72	0.71	0.80	0.64	0.75	0.65	0.74	0.51	0.35	0.60	0.44	0.59	0.46
	SVR	0.70	0.63	0.71	0.64	0.70	0.60	0.72	0.79	0.70	0.77	0.72	0.81	0.49	0.38	0.52	0.41	0.49	0.35

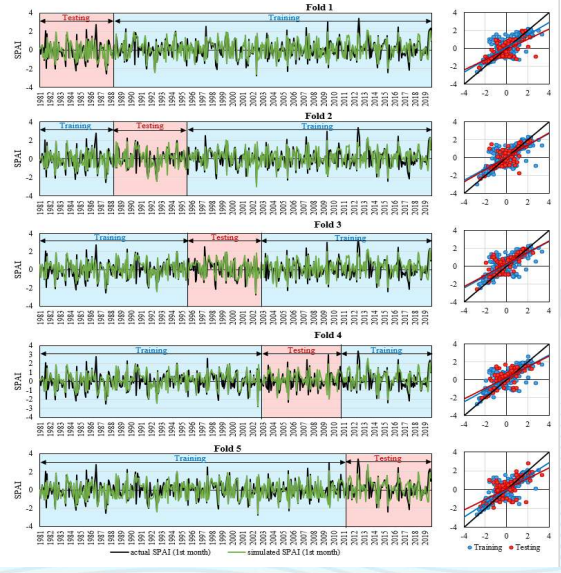
Results and Discussions



Model performance in drought identification for TRB study basin using DL based model.

- Observed and simulated (1st month) SPAI are shown through time series (left) and scatter plots (right) for all 5 folds.
- In the scatter plots, the x-axis and y-axis represent the observed and simulated SPAI values, respectively.

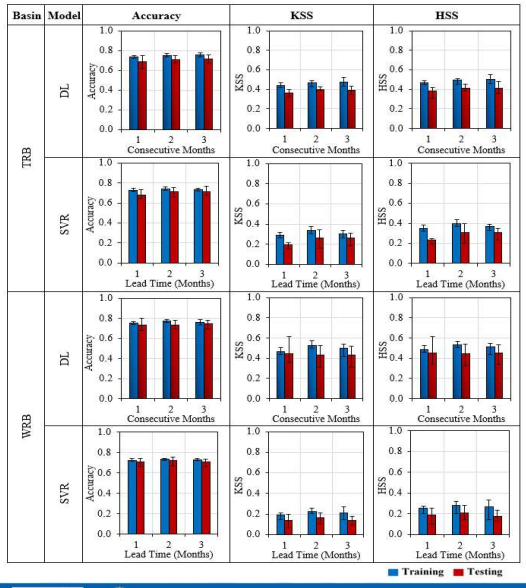
Results and Discussions



Model performance in drought identification for WRB study basin using DL based model.

- Observed and simulated (1st month) SPAI are shown through time series (left) and scatter plots (right) for all 5 folds.
- In the scatter plots, the x-axis and y-axis represent the observed and simulated SPAI values, respectively.

Results and Discussions



- Average *accuracy*, *KSS* and *HSS*,
- Range bars at the top show the maximum to minimum value, across different folds for drought category identification for both the study basins using Deep Learning (DL) and Support Vector Regression (SVR) models for 1st, 2nd and 3rd months.

Results and Discussions

Fold	Perform. measures	TRB						WRB					
		1st month		2nd month		3rd month		1st month		2nd month		3rd month	
		trn	tst	trn	tst	trn	tst	trn	tst	trn	tst	trn	tst
Fold 1	accuracy	**0.73** 0.73	**0.75** 0.70	**0.74** 0.73	**0.74** 0.75	**0.75** 0.72	**0.73** 0.74	**0.75** 0.74	**0.75** 0.66	**0.79** 0.75	**0.69** 0.67	**0.79** 0.75	**0.69** 0.66
	KSS	**0.43** 0.32	**0.40** 0.19	**0.47** 0.33	**0.39** 0.31	**0.50** 0.32	**0.40** 0.31	**0.46** 0.19	**0.45** 0.06	**0.57** 0.22	**0.37** 0.11	**0.54** 0.21	**0.36** 0.09
	HSS	**0.46** 0.39	**0.42** 0.23	**0.49** 0.40	**0.40** 0.35	**0.51** 0.38	**0.40** 0.35	**0.47** 0.27	**0.50** 0.10	**0.57** 0.29	**0.40** 0.15	**0.55** 0.28	**0.40** 0.13
Fold 2	accuracy	**0.74** 0.75	**0.71** 0.74	**0.76** 0.76	**0.75** 0.76	**0.76** 0.75	**0.76** 0.77	**0.74** 0.72	**0.80** 0.71	**0.76** 0.74	**0.78** 0.73	**0.76** 0.72	**0.78** 0.71
	KSS	**0.44** 0.29	**0.38** 0.22	**0.48** 0.31	**0.42** 0.23	**0.50** 0.29	**0.44** 0.24	**0.44** 0.15	**0.61** 0.16	**0.50** 0.20	**0.52** 0.21	**0.50** 0.14	**0.49** 0.15
	HSS	**0.47** 0.36	**0.40** 0.25	**0.51** 0.38	**0.46** 0.28	**0.52** 0.35	**0.48** 0.30	**0.46** 0.21	**0.61** 0.23	**0.51** 0.20	**0.54** 0.21	**0.51** 0.14	**0.53** 0.15
Fold 3	accuracy	**0.72** 0.71	**0.62** 0.65	**0.73** 0.74	**0.67** 0.66	**0.73** 0.73	**0.67** 0.69	**0.76** 0.73	**0.74** 0.70	**0.76** 0.74	**0.76** 0.73	**0.73** 0.74	**0.77** 0.71
	KSS	**0.42** 0.31	**0.32** 0.15	**0.43** 0.35	**0.39** 0.16	**0.44** 0.34	**0.36** 0.19	**0.48** 0.21	**0.41** 0.19	**0.51** 0.25	**0.49** 0.21	**0.42** 0.27	**0.50** 0.12
	HSS	**0.44** 0.36	**0.31** 0.19	**0.45** 0.41	**0.38** 0.20	**0.46** 0.39	**0.36** 0.24	**0.50** 0.28	**0.45** 0.26	**0.51** 0.32	**0.51** 0.28	**0.44** 0.34	**0.52** 0.18
Fold 4	accuracy	**0.74** 0.72	**0.67** 0.65	**0.74** 0.72	**0.67** 0.65	**0.78** 0.74	**0.70** 0.68	**0.74** 0.72	**0.71** 0.74	**0.74** 0.72	**0.71** 0.74	**0.77** 0.72	**0.76** 0.73
	KSS	**0.47** 0.27	**0.34** 0.21	**0.49** 0.38	**0.41** 0.26	**0.52** 0.32	**0.41** 0.28	**0.45** 0.20	**0.38** 0.10	**0.52** 0.22	**0.48** 0.11	**0.53** 0.23	**0.52** 0.14
	HSS	**0.49** 0.33	**0.36** 0.25	**0.51** 0.44	**0.42** 0.31	**0.55** 0.39	**0.43** 0.33	**0.48** 0.27	**0.37** 0.14	**0.55** 0.29	**0.46** 0.15	**0.54** 0.30	**0.49** 0.18
Fold 5	accuracy	**0.75** 0.72	**0.70** 0.65	**0.77** 0.73	**0.71** 0.71	**0.75** 0.72	**0.70** 0.68	**0.77** 0.72	**0.68** 0.74	**0.77** 0.73	**0.70** 0.76	**0.77** 0.72	**0.72** 0.73
	KSS	**0.45** 0.26	**0.38** 0.21	**0.47** 0.31	**0.39** 0.34	**0.44** 0.26	**0.35** 0.29	**0.51** 0.20	**0.36** 0.17	**0.52** 0.23	**0.31** 0.19	**0.51** 0.21	**0.31** 0.18
	HSS	**0.48** 0.33	**0.42** 0.25	**0.51** 0.37	**0.43** 0.40	**0.47** 0.33	**0.39** 0.33	**0.53** 0.27	**0.34** 0.23	**0.54** 0.31	**0.33** 0.27	**0.53** 0.28	**0.34** 0.23
Average	accuracy	**0.74** 0.72	**0.69** 0.68	**0.75** 0.74	**0.71** 0.71	**0.75** 0.73	**0.71** 0.71	**0.75** 0.73	**0.74** 0.71	**0.77** 0.73	**0.73** 0.72	**0.76** 0.73	**0.74** 0.71
	KSS	**0.44** 0.29	**0.36** 0.20	**0.47** 0.33	**0.40** 0.26	**0.48** 0.31	**0.39** 0.26	**0.47** 0.19	**0.44** 0.14	**0.53** 0.23	**0.44** 0.17	**0.50** 0.21	**0.43** 0.14
	HSS	**0.47** 0.35	**0.38** 0.23	**0.50** 0.40	**0.42** 0.31	**0.50** 0.37	**0.41** 0.31	**0.49** 0.26	**0.45** 0.19	**0.54** 0.28	**0.45** 0.21	**0.51** 0.27	**0.45** 0.18

Fold-wise performance measures between observed and simulated drought category

- training (trn) and testing (tst) period for 1st, 2nd and 3rd months for both the study basins (TRB and WRB).
- Each cell shows values for DL based model (top and bold) and SVR (bottom).

Results and Discussions

TRB

Drought category		Simulated							
		D2	D1	D0	N	W0	W1	W2	Sum
Observed	D2	2	5	1	1	0	0	0	9
	D1	1	12	7	5	0	0	0	25
	D0	1	2	9	25	0	1	0	38
	N	0	1	15	264	19	9	0	308
	W0	0	0	0	23	17	3	1	44
	W1	0	0	0	6	5	10	3	24
	W2	0	0	0	1	2	5	2	10
	Sum	4	20	32	325	43	28	6	458

WRB

Drought category		Simulated							
		D2	D1	D0	N	W0	W1	W2	Sum
Observed	D2	5	1	3	1	0	0	0	10
	D1	0	3	8	6	1	0	0	18
	D0	0	2	12	21	3	0	0	38
	N	0	0	12	285	15	4	0	316
	W0	0	1	1	12	16	10	0	40
	W1	0	0	1	0	7	10	1	19
	W2	0	0	0	2	5	9	1	17
	Sum	5	7	37	327	47	33	2	458

- Contingency table for the observed and simulated drought categories using DL based model for both study basin considering all the testing periods in different folds so as to consider the entire time period of analysis as testing period only.

CONCLUSIONS

- Conv1D successfully captured the complex relationship between hydrometeorological precursors and rainfall variability for basin-scale drought assessment.
- The proposed DL model generally outperformed SVR in both continuous SPAI simulation and categorical drought assessment.
- Conv1D could generate outputs for three consecutive months simultaneously, whereas SVR required separate training for individual months.
- The framework demonstrates the potential to use comparatively better-simulated primary meteorological variables from climate models for assessment of more complex phenomena such as drought.
- A major limitation is the large data requirement for Deep Learning; spatial transferability to data-scarce regions was identified as a future research direction.


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Thank You!



CONCEPTS COVERED

Application of Deep Learning to Explore Future Streamflow Characteristics under Climate Change

NPTEL

Introduction

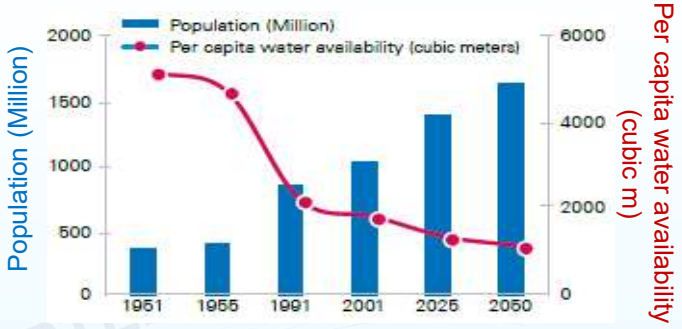
- Access to safe water, sanitation and hygiene is the basic human need for health and well-being, and hence, been considered as one of the seventeen Sustainable Development Goals (SDGs) aimed by the United Nations (UN, 2016).
- Almost two-thirds of the world's population experiences severe water scarcity for at least one month each year.



Source: https://education.nationalgeographic.org/resource/sustainable-development-goals/

Introduction

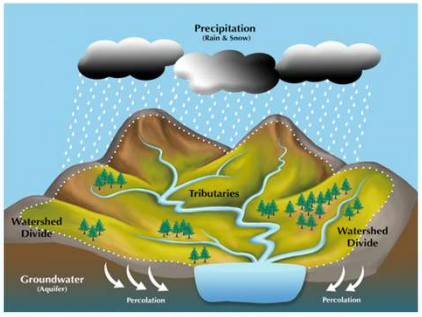
- Demand for water is definitely rising worldwide including India, owing to rapid population growth, urbanization and increasing water needs from agriculture, industry, and energy sectors.
- As a result, the per capita water availability is continuously dropping over last few decades.



Source: Ministry of Water Resources, Govt. of India

Introduction

- The situation gets further exacerbated due to inefficient use of water and the present scenario of changing climate.
- In this context, streamflow being an important component of the global hydrological cycle and having a significant share in freshwater availability for drinking, agricultural and industrial purpose, its efficient modelling and reliable long-term projections are of utmost importance.
- Several General Circulation Models (GCMs) are used for future projection of climate variables. The simulations results are available for the entire world.



Source: https://www.seacoastsciencecenter.org/learning-resource/whats-a-watershed/

Motivation of the Study

- Though the GCMs are able to simulate the primary hydrometeorological variables (e.g., temperature, precipitation etc.) with reasonably good accuracy, their performance in simulating secondary variables (e.g., streamflow), or tertiary variables (e.g., as drought) is still inferior.
- Reliable modelling of the association between a set of primary hydrometeorological variables and streamflow will be helpful to beneficially translate the potential of GCMs for an improved simulation of streamflow, and subsequently for its reliable future assessment.
- Given the inherent complexity of this problem, machine learning approaches offer the potential to reveal valuable insights. This builds the motivation behind the study.

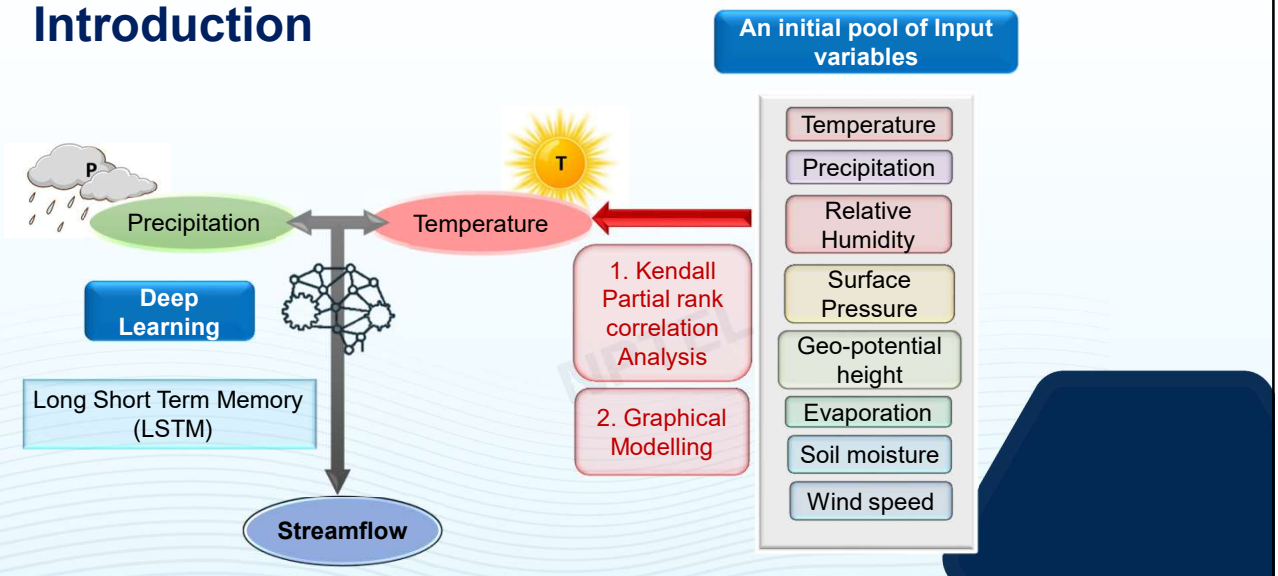
Objective of the Study

- The overall objective of the study is to enhance the usability of GCM simulations for future assessment of basin-scale streamflow using Deep Learning.
  - Two specific points to discuss in this lecture:
    - Developing an LSTM model capable of capturing the hidden complex association between streamflow and its hydrometeorological precursors.
    - Assessing anticipated future streamflow variability under SSP2-4.5 and SSP5-8.5 climate-change scenarios.

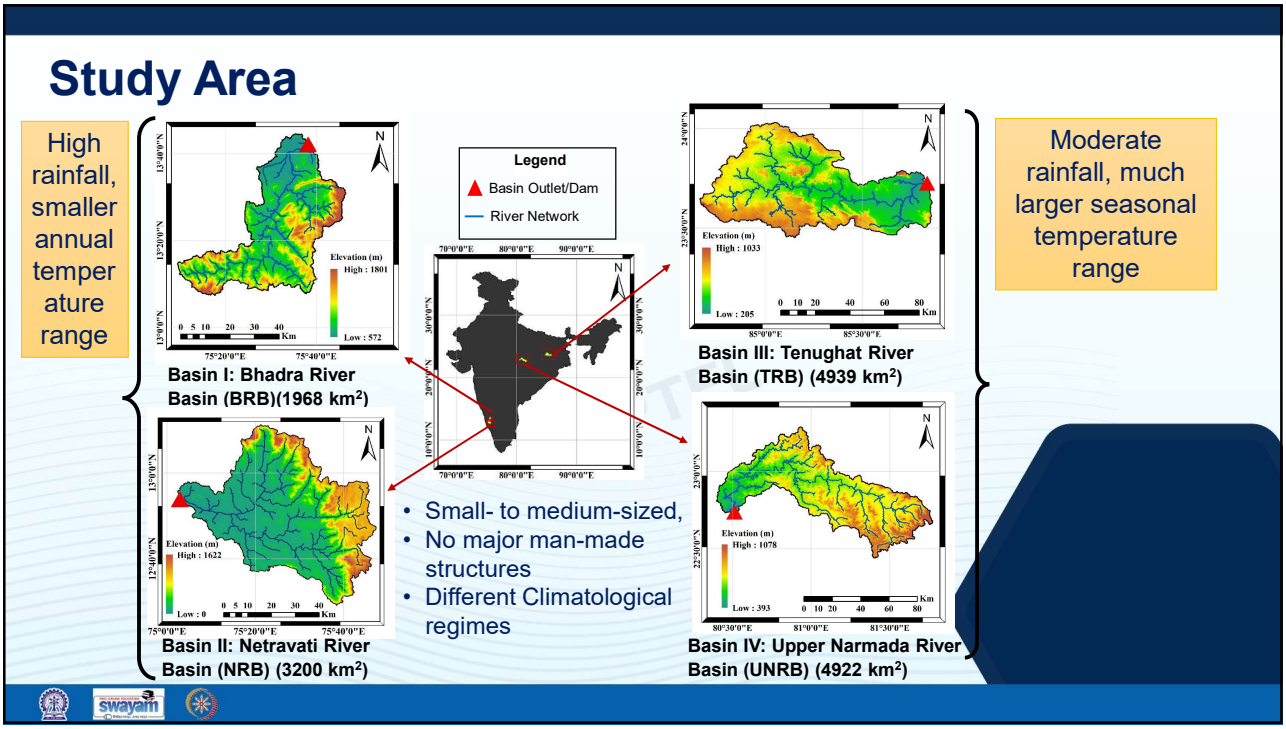
Shared Socioeconomic Pathway (SSP)

- These are standardized climate change scenario used by the Intergovernmental Panel on Climate Change (IPCC) in its Sixth Assessment Report (AR6) and the Coupled Model Intercomparison Project Phase 6 (CMIP6).
- SSP2-4.5 represents an “intermediate” or “middle-of-the-road” trajectory for future greenhouse gas emissions and global socioeconomic development.
- SSP5-8.5 represents a future characterized by rapid global economic growth, energy-intensive lifestyles, and a continued heavy reliance on fossil fuels, leading to severe atmospheric changes by the end of the century.
- There are other pathways, such as SSP1-1.9, SSP1-2.6 and SSP3-7.0. Details can be found from AR6 as mentioned before.

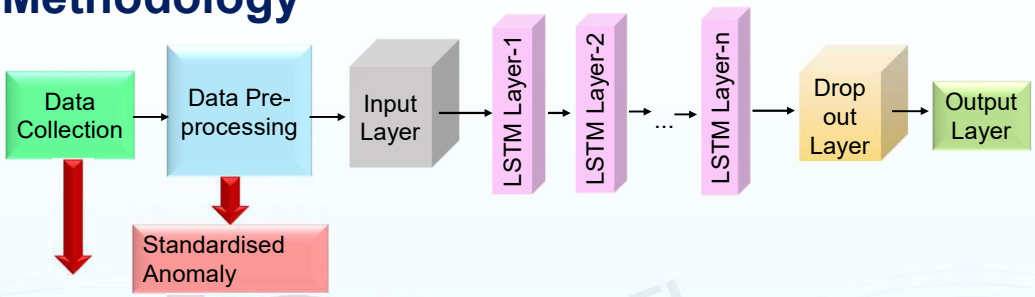
Introduction



Study Area

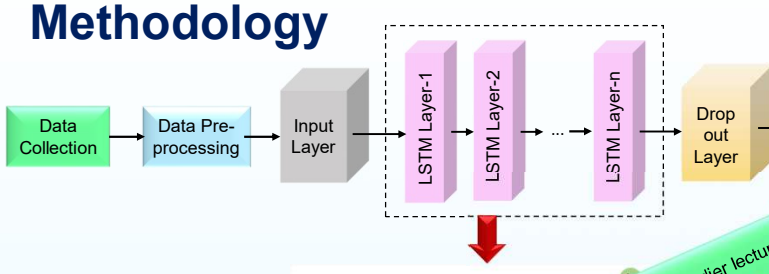


Methodology



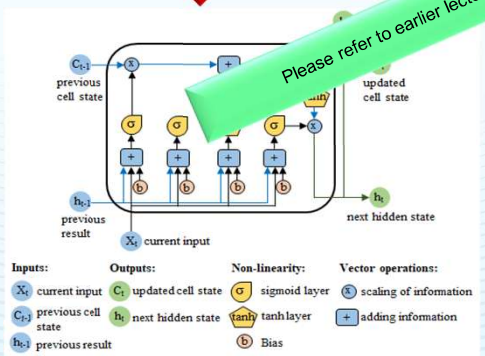
Data Type	Variable Name	Source
Input Data / Causal Variables	Precipitation (P) and Temperature (T)	• India Meteorological Department (IMD)
Output data	Monthly Streamflow (Q)	• BRB: Karnataka Niravari Nigam Limited (KNNL) • TRB: Damodar Valley Corporation (DVC) • NRB, UNRB: Water Resource Information System (WRIS), India

Methodology





Please refer to earlier lectures



Forget gate: $f_t = \sigma(W_f.[h_{t-1}, X_t] + b_f)$

$i_t = \sigma(W_i.[h_{t-1}, X_t] + b_i)$

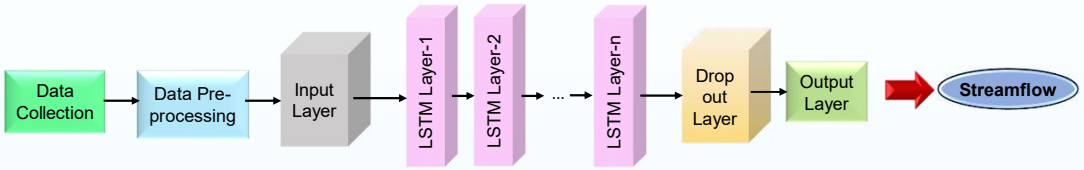
$\tilde{C}_t = tanh(W_c[h_{t-1}, X_t] + b_c)$

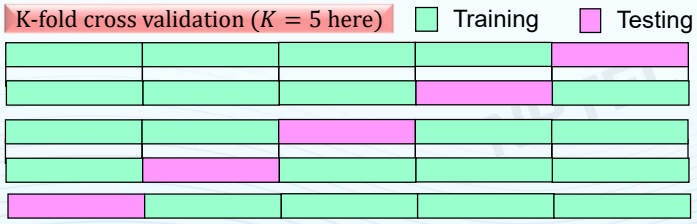
Cell state: $C_t = i_t * \tilde{C}_t + f_t * C_{t-1}$

Output gate: $o_t = \sigma(W_0[h_{t-1}, X_t] + b_0)$

New hidden state: $h_t = o_t * tanh(C_t)$

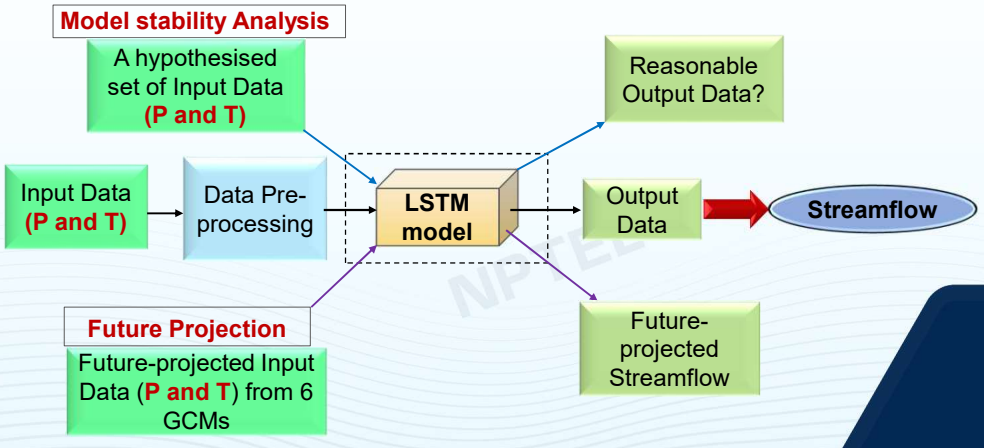
Methodology





- The entire dataset is divided into K approximately equal and disjoint folds, and the model is recurrently trained on remaining $(K - 1)$ folds, and its performance is measured on the remaining one fold.

Methodology



Results

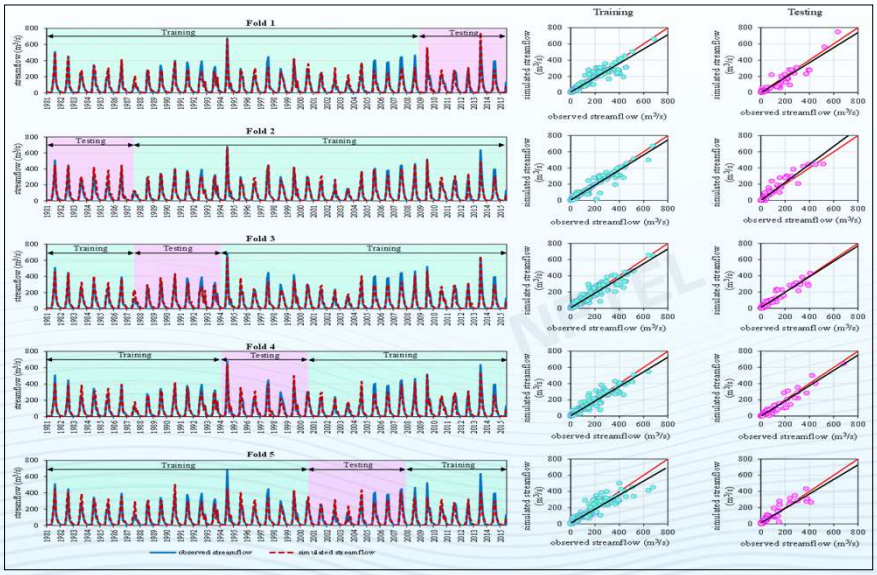
- All performance measures indicate a good performance of the LSTM model in capturing the monthly streamflow variation in all four basins.

Basins	Selected optimized hyperparameters of the LSTM model					Fold-averaged model performance in training (testing) period		
	No. of LSTM layers	Batch size	LSTM unit	Epoch	Learning rate	CC	RMSE (m^3/s)	NSE
BRB	12	48	20	3000	0.0001	0.96 (0.95)	34.27 (38.52)	0.92 (0.89)
NRB	11	66	56	500	0.001	0.98 (0.98)	103.21 (109.40)	0.95 (0.94)
TRB	10	12	20	1500	0.0001	0.94 (0.92)	37.97 (43.78)	0.88 (0.83)
UNRB	10	330	20	1000	0.001	0.96 (0.92)	46.61 (64.53)	0.92 (0.85)

- Also, the performances of the training and testing period remain comparable for all folds, ensuring optimal training of the model (no overfitting/underfitting).

NPTEL

Results (for BRB)



Model Stability and Sensitivity Analysis

BRB	Temperature Changes (°C)						
Precipitation Changes (%)	-1	-0.5	0	0.5	1	1.5	2
-50	-1.83	-10.74	-24.48	-32.04	-31.88	-27.67	-33.19
-25	11.58	8.77	-12.21	-21.95	-26.80	-30.76	-31.56
0	29.30	12.33	0.00	-11.20	-19.95	-26.04	-27.49
25	40.79	27.06	12.19	0.10	-6.93	-13.07	-20.68
50	50.18	31.71	25.24	12.58	1.53	-3.30	-8.24
75	61.74	51.45	34.05	26.65	18.29	9.36	0.96
100	79.51	65.12	45.03	38.18	31.56	22.74	15.58

NRB	Temperature Changes (°C)						
Precipitation Changes (%)	-1	-0.5	0	0.5	1	1.5	2
-50	-16.09	-23.29	-24.81	-30.65	-33.75	-35.43	-36.00
-25	2.11	-7.07	-13.27	-16.28	-20.01	-22.16	-24.41
0	16.90	9.82	0.00	-5.64	-18.43	-19.86	-22.54
25	31.88	24.66	13.95	6.36	-9.82	-15.45	-20.55
50	36.35	31.81	24.23	10.31	-4.75	-8.43	-10.84
75	45.71	43.10	34.25	16.69	10.96	8.50	3.95
100	63.53	47.74	35.74	29.50	21.93	15.02	11.95

TRB	Temperature Changes (°C)						
Precipitation Changes (%)	-1	-0.5	0	0.5	1	1.5	2
-50	-3.22	-9.72	-23.23	-38.05	-46.51	-48.36	-40.23
-25	4.72	-9.05	-17.92	-25.66	-33.57	-38.34	-47.01
0	10.17	7.68	0.00	-5.23	-16.79	-26.08	-36.45
25	16.48	25.80	20.65	7.65	-2.10	-12.01	-19.70
50	30.06	31.76	34.58	24.87	11.50	1.92	-9.94
75	31.45	33.66	33.67	32.53	27.24	13.43	4.76
100	33.04	38.16	36.09	31.77	28.65	21.36	14.92

UNRB	Temperature Changes (°C)						
Precipitation Changes (%)	-1	-0.5	0	0.5	1	1.5	2
-50	1.84	-17.79	-32.00	-44.79	-47.47	-57.38	-57.34
-25	5.01	-6.85	-11.33	-29.38	-38.90	-38.08	-47.04
0	12.64	11.64	0.00	-6.54	-26.04	-23.33	-35.93
25	22.23	20.73	5.46	-4.78	-9.69	-15.42	-18.04
50	25.89	24.36	17.61	10.60	4.56	3.14	2.11
75	29.78	34.73	14.60	15.04	14.10	10.68	6.40
100	31.86	39.33	23.14	18.03	12.48	12.16	8.82

- To check the stability, a hypothesised set of 49 experiments is devised to stress the proposed LSTM model enough beyond its usual range of possible inputs.
- Results show realistic yet complex and non-linear association between streamflow and hydrometeorological precursors.

Model Stability and Sensitivity Analysis

BRB	Temperature Changes (°C)						
Precipitation Changes (%)	-1	-0.5	0	0.5	1	1.5	2
-50	-1.83	-10.74	-24.48	-32.04	-31.88	-27.67	-33.19
-25	11.58	8.77	-12.21	-21.95	-26.80	-30.76	-31.56
0	29.30	12.33	0.00	-11.20	-19.95	-26.04	-27.49
25	40.79	27.06	12.19	0.10	-6.93	-13.07	-20.68
50	50.18	31.71	25.24	12.58	1.53	-3.30	-8.24
75	61.74	51.45	34.05	26.65	18.29	9.36	0.96
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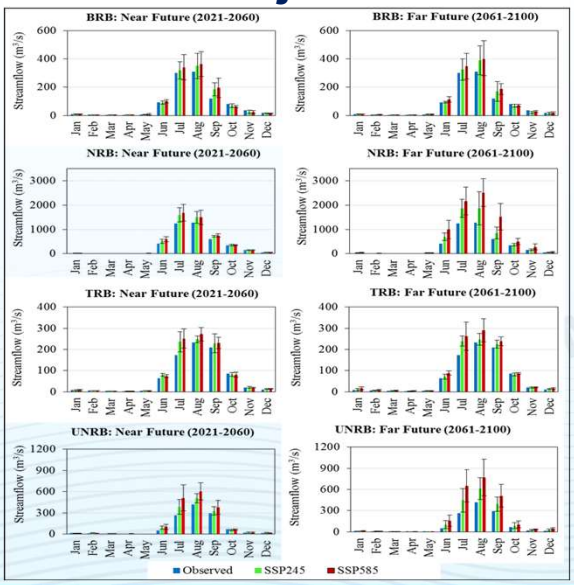
NRB	Temperature Changes (°C)						
Precipitation Changes (%)	-1	-0.5	0	0.5	1	1.5	2
-50	-16.09	-23.29	-24.81	-30.65	-33.75	-35.43	-36.00
-25	2.11	-7.07	-13.27	-16.28	-20.01	-22.16	-24.41
0	16.90	9.82	0.00	-5.64	-18.43	-19.86	-22.54
25	31.88	24.66	13.95	6.36	-9.82	-15.45	-20.55
50	36.35	31.81	24.23	10.31	-4.75	-8.43	-10.84
75	45.71	43.10	34.25	16.69	10.96	8.50	3.95
100	63.53	47.74	35.74	29.50	21.93	15.02	11.95

TRB	Temperature Changes (°C)						
Precipitation Changes (%)	-1	-0.5	0	0.5	1	1.5	2
-50	-3.22	-9.72	-23.23	-38.05	-46.51	-48.36	-40.23
-25	4.72	-9.05	-17.92	-25.66	-33.57	-38.34	-47.01
0	10.17	7.68	0.00	-5.23	-16.79	-26.08	-36.45
25	16.48	25.80	20.65	7.65	-2.10	-12.01	-19.70
50	30.06	31.76	34.58	24.87	11.50	1.92	-9.94
75	31.45	33.66	33.67	32.53	27.24	13.43	4.76
100	33.04	38.16	36.09	31.77	28.65	21.36	14.92

UNRB	Temperature Changes (°C)						
Precipitation Changes (%)	-1	-0.5	0	0.5	1	1.5	2
-50	1.84	-17.79	-32.00	-44.79	-47.47	-57.38	-57.34
-25	5.01	-6.85	-11.33	-29.38	-38.90	-38.08	-47.04
0	12.64	11.64	0.00	-6.54	-26.04	-23.33	-35.93
25	22.23	20.73	5.46	-4.78	-9.69	-15.42	-18.04
50	25.89	24.36	17.61	10.60	4.56	3.14	2.11
75	29.78	34.73	14.60	15.04	14.10	10.68	6.40
100	31.86	39.33	23.14	18.03	12.48	12.16	8.82

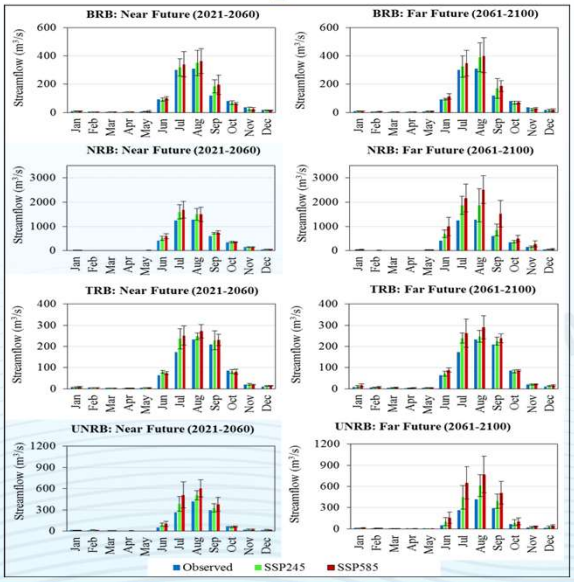
- In general, the streamflow is found to depict the logical negative association with temperature and positive association with precipitation.

Future Projection of Streamflow



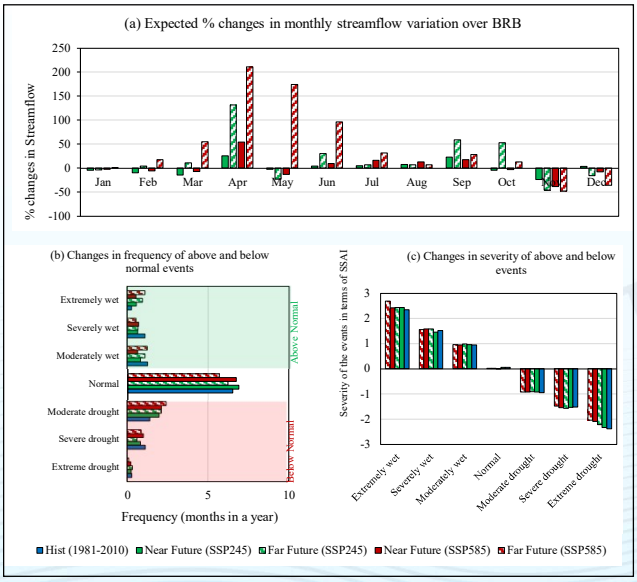
- For future streamflow assessment, a future-projected bias-corrected dataset of Precipitation and Temperature values from six CMIP6 GCMs are used for two SSPs, namely SSP245 and SSP585.
- The monsoon months (i.e. the climatological high-streamflow months) are expected to have even more streamflow for all four basins.
- Streamflow during the low-flow months more-or-less remains the same or declines further over time.

Future Projection of Streamflow



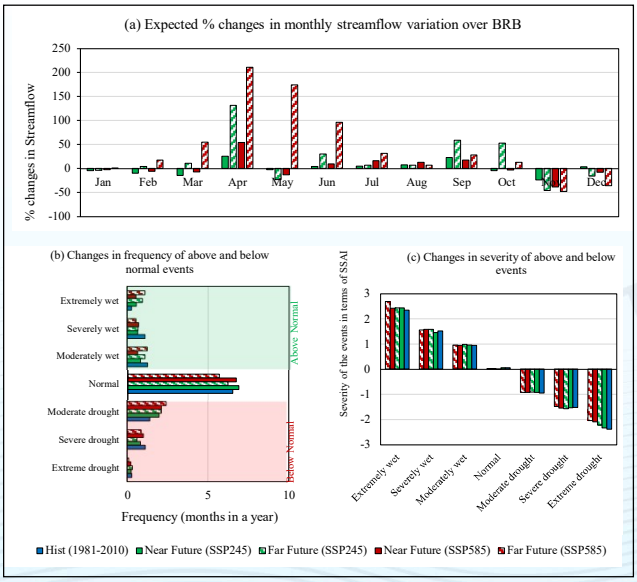
- Moreover, it is evident that the extent of decline in low-flow months and rise in high-flow months increases in the far-future period compared to the near-future.
- Higher risks of floods and droughts both.
- Magnitude of the increase or decrease in flow is greater under SSP5-8.5 as compared to SSP2-4.5.

Future Projection of Streamflow



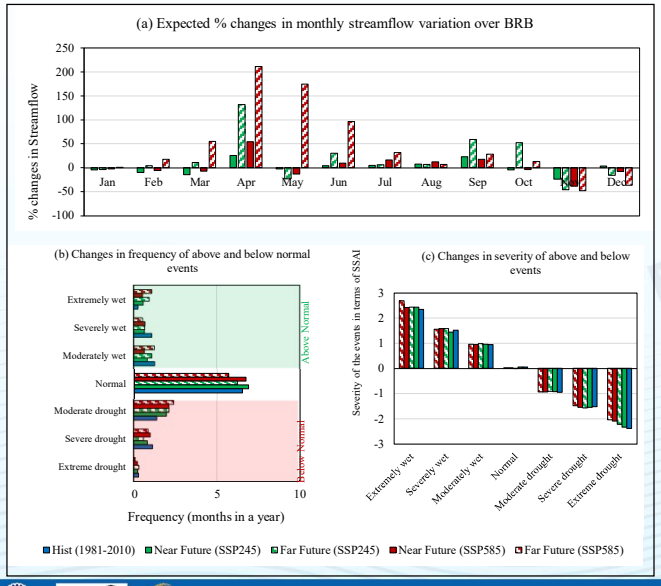
- To explore this further, we estimated the month-wise changes (in percentage) in future-projected streamflow over the future with reference to the 1981-2010 baseline. Results are presented here for BRB.
- From this plot, it can be seen that the flow is expected to reduce for the low–flow months such as November through March, after which it shows some rise, particularly for pre-monsoon and monsoon months.

Future Projection of Streamflow



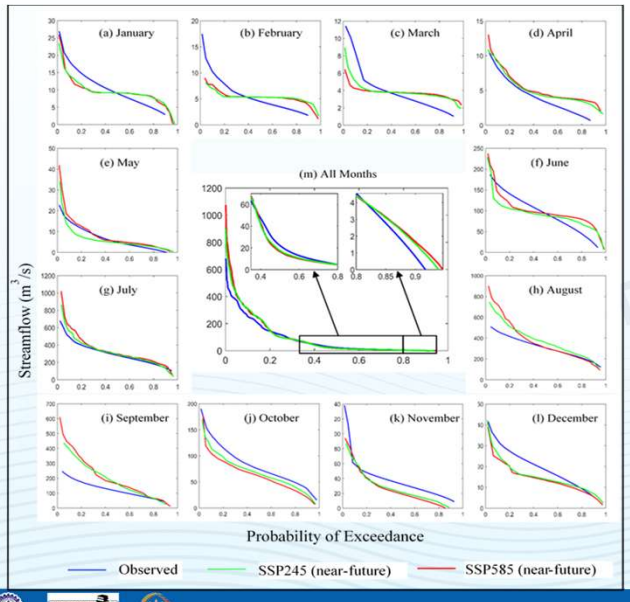
- Next, to explore the potential risk of floods and droughts in the study basins, we considered Standardized Streamflow Anomaly Index (SSAI).
- Results indicate a substantial increase for extremely wet events, both in terms of frequency and its severity.
- Although the frequency of moderate and severe wet events will not increase in the future, their severity is increasing, even though marginal.

Future Projection of Streamflow



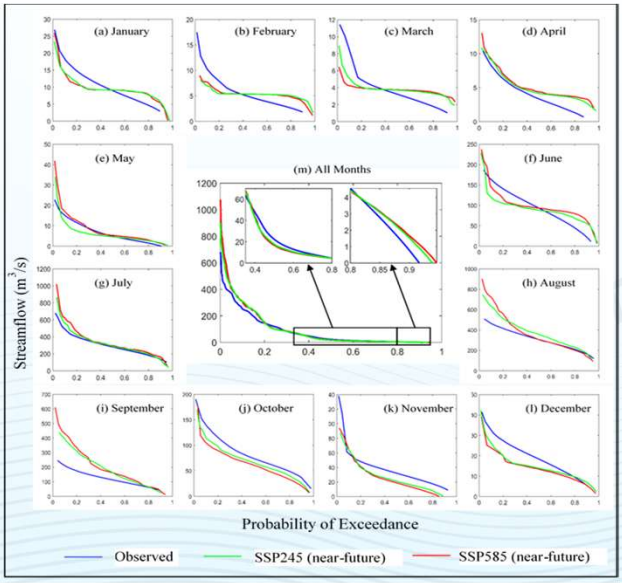
- On the contrary, moderate drought events are expected to increase in future, whereas extreme droughts exhibit a reducing trend.
- Overall, we expect the risk of potential flood and drought – both may increase under the changing climate over future.

Future Flow-Duration Curves (FDCs)



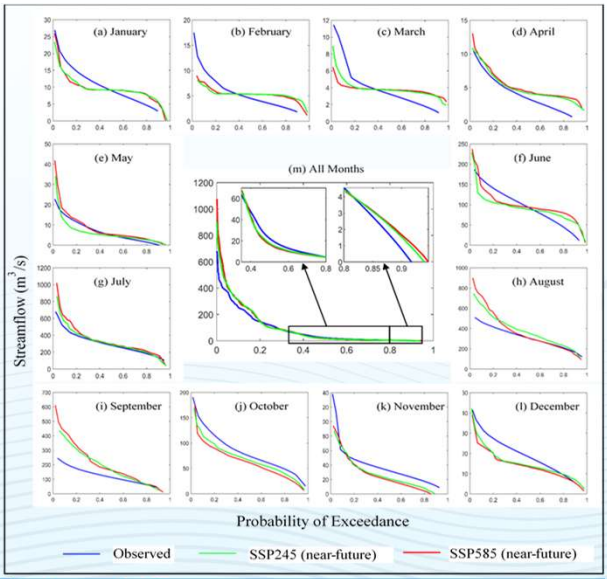
- Future Flow-Duration Curves are also obtained for all the moths.
- Plots are shown for Bhadra River Basin (BRB) for the near-future period.
- The high flow events (approx. Above 70th percentile) are going to be more probable.

Future Flow-Duration Curves (FDCs)



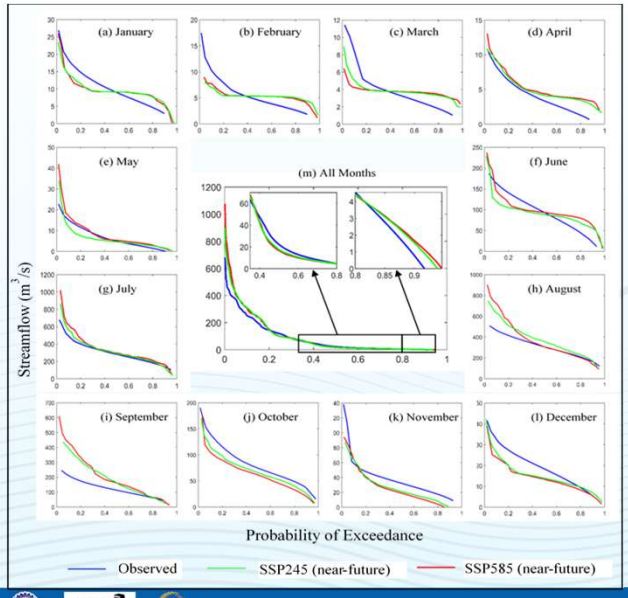
- The moderate range of streamflow (20th to 70th percentile), the observation is opposite, i.e., the projected curves are below the observed FDC, indicating a potential reduction in their probability of occurrences in near-future.

Future Flow-Duration Curves (FDCs)



- Again, towards the right-most tail of the curves, i.e., for the lower magnitudes of streamflow (approximately below 20th percentile), the projected curves are placed above the observed one, indicating an increase in low streamflow regimes as well.
- That means both ends of the streamflow regime, i.e., the higher as well as the lower extremes are expected to become more probable or frequent as a potential impact of climate change.

Future Flow-Duration Curves (FDCs)



- This observation is further strengthened by the FDCs for individual months, which clearly shows a reduction (rise) in high flow (low flow) in the climatologically dry months (October to march).
- Since April, the increase in high flow is gradually observed which attains a maximum level in monsoon months such as July, August, and September.

CONCLUSIONS

- This study explore the potential of DL by proposing a novel LSTM architecture to model monthly streamflow using two hydrometeorological precursors – precipitation and temperature.
- The stability and sensitivity of the developed model are thoroughly checked by 49 combinations of hypothesized hydrometeorological forcings.
- After ensuring stability and efficient performance of the developed model, it is used for multi-model long-term projection of streamflow across all four basins.
- The results reveal a couple of interesting findings. Particularly, it indicates an increased streamflow under a wetter and hotter climate in future, which gets more pronounced with time and under the higher forcing scenario.

Reference:
Sarkar, S., Khan, M. I., and Maity, R. (2025). Deep learning reveals future streamflow characteristics change and climate sensitivity. Journal of Hydrology, 660, 133457.


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Thank You!